

Deep Learning-Based Emotion Recognition using EEG: A Comprehensive Review

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Abstract

Emotion recognition using EEG is a growing trend in affective computing, with highly fragmented studies regarding datasets, approaches, and evaluation metrics, as well as a high degree of inter-individual variability and a rapidly changing set of deep learning approaches. In this scenario, it is difficult to extract a comprehensive view on the generalizability and practicability of the approaches considered. This work offers a structured review of emotion recognition using EEG, aiming to decrease this level of fragmentation by grouping the existing approaches following signal processing, representation learning, deep learning architectures, and cross-subject learning strategies. The most widely adopted deep learning architectures, graph-based, recurrent, and convolutional neural networks, are considered regarding their set of architectural choices, strengths, and weaknesses, as well as related trade-offs. Other major issues, overfitting, small dataset, and subject dependency, are illustrated and critically considered together with more recent approaches trying to increase the reliability of EEG emotion recognition systems. According to recent studies, the accuracy achieved in subject-specific approaches is higher than 95%, while for the subject-independent approach, accuracy values vary from 70% to 90%, depending on the dataset and the architecture considered.

1 Introduction

Emotion recognition is at the heart of models of human behavior, since emotional states directly and profoundly affect human cognition, decision-making, and social interactions. Automatic emotion recognition facilitates the development of intelligent systems capable of adaptively responding to the affect states of users, and this is particularly critical in contemporary human-computer interaction tasks. Emotion recognition in psychology and neuroscience helps in the objective analysis of affective phenomena independently of subjective self-assessments, and in the health sector, it paves the way for numerous tasks, such as the objective evaluation of mood disturbances, patient recovery, and supportive treatment of neurodevelopmental disorders such as autism and affect dysregulation. Emotion recognition in engineering and brain-computer interface studies helps in developing adaptive systems capable of automatically responding to the affective states of users. Emotion recognition using EEG recordings is particularly useful in patient-oriented tasks in the health sector, such as patient recovery and long-term evaluation of patient moods, wherein overt behavioral manifestations of patients could be ambiguous and unavailing. EEG emotion recognition stands distinct, as it directly taps brain signals related to emotion recognition, and this makes the technology valuable in tasks where robustness and physiological validity of the detection tasks are critical.

Electroencephalography (EEG) is a non-invasive neurophysiological sensing technique that records electrical activity generated by neuronal populations with high temporal resolution. Because EEG captures

brain dynamics at the millisecond level, it provides fine-grained insight into neural processes underlying emotional responses, which are often transient and rapidly evolving [1, 2]. EEG-based emotion recognition systems exploit this close coupling between brain activity and affective processing by analyzing neural responses elicited during emotional stimulation to infer underlying emotional states [3]. This neurophysiological grounding distinguishes EEG from peripheral or behavioral modalities and motivates its widespread adoption in affective computing research.

EEG further provides important advantages compared to other emotion recognition methods, like facial expressions, speech, or body gestures. Behavioral cues are susceptible to intentional control, social camouflage, and context effects and are thus not very reliable in a sensitive context. Conversely, EEG oscillations are symptomatic of inner neural activities that are hard to manipulate intentionally and supply objective estimations of affective states [4]. Although increasingly explored, at least for the future of emotion-sensitive BCIs, multimodal approaches combining EEG with other physiological or behavioral signals are not required, and EEG alone can be a robust and informative modality owing to its direct sensitivity to emotional brain dynamics and the compatibility with BCI systems.

Despite having all these benefits, there are a few challenges to recognizing emotions using EEG. The quality of an EEG signal is highly dependent on factors like electrode positioning, skin conductance, movement artifacts, and inter-session inconsistency. Variability in experimental procedures, emotional stimuli used, labeling approaches, and preprocessing techniques adds to inconsistencies [5]. These create difficulties in comparing methodology-based approaches systematically, especially when assessing performance on diverse databases such as DEAP, SEED, and DREAMER [2, 6]. Additionally, large variations between subjects lead to large distribution changes in the features of an EEG signal, thereby negatively affecting subject-independent generalization.

Recent breakthroughs in deep learning have led to significant improvements in EEG-based affect recognition models that are capable of learning informative features directly from raw or lightly preprocessed signals [4]. Convolutional Neural Networks (CNNs) capture channel-level relations in EEG signals [7], Recurrent Neural Networks (RNNs) model temporal dependencies in brain activity dynamics [8], and Graph Neural Networks (GNNs) represent task-relevant functional connectivity between brain regions [9]. Despite their strong ability in achieving a high level of accurate classification for subject-dependent tasks, they have been observed to have poor generalization capability for cross-subject and cross-corpus [10].

There has been an increased need to carry out a review of up-to-date research work in the area of emotion recognition using EEG. There are many reviews that were mainly concentrating on architectural originality or evaluation tasks on a single dataset, with less concern for cross-subject and cross-dataset generalization [5, 4]. Additionally, a lack of standardized preprocessing pipelines and inconsistencies in evaluation methodologies make it hard to make a valid inference [11] on the research. There is no consensus or agreement on the best way for model design, evaluation, and implementation of effective EEG-based emotion recognition [12].

1.1 Contributions

Compared with recent surveys and reviews on EEG-based emotion recognition and deep learning this review contributes:

1. **End-to-end coverage from theory to practice:** a unified narrative that connects emotion theory (discrete + dimensional models), neurophysiological foundations, stimulus/induction design, acquisition hardware evolution, multimodal affect modeling, and EEG correlates to the complete learning pipeline (preprocessing → feature extraction → feature selection → modeling).
2. **A section-aligned taxonomy matching real research workflows:** a structured map that mirrors how EEG-ER systems are built (data/design → sensing → processing → modeling → evaluation), making it easier to locate open problems and best practices than method-only taxonomies.
3. **Cross-domain feature viewpoint consolidated into one framework:** explicit, side-by-side

synthesis of temporal, frequency, spatial, temporal–frequency, and fusion-domain feature concepts, including how modern deep models operationalize these representations.

4. **Evaluation rigor emphasized beyond “accuracy”:** clearer separation of subject-dependent vs. subject-independent protocols and how they affect reported performance, plus a consolidated view of metrics, dataset splits, and reproducibility pitfalls.
5. **Challenges-to-strategies linking:** each major challenge (redundancy, scarcity, cross-subject generalization, practical deployment constraints) is paired with concrete strategy families (selection/regularization/augmentation/generation, domain adaptation, lightweight pipelines).
6. **Modern comparison lens:** an explicit “what is covered vs. missing” matrix across leading recent reviews, highlighting under-covered but important aspects (stimulus design, acquisition systems, multimodal modeling, and practical constraints) that this paper includes.

Table 1: Comparison of coverage across recent EEG emotion recognition reviews. The proposed review (This Work) provides comprehensive coverage across all major dimensions.

Attribute	Jafari 2023 [13]	Wang 2024 [14]	Ma 2024 [15]	Erat 2024 [16]	Chen 2025 [17]	Geng 2025 [18]	This Work
Emotion-Theory		✓					✓
Subject-Dependent		✓	✓			✓	✓
Subject-Independent		✓	✓			✓	✓
Acquisition							✓
Signal Processing	✓	✓	✓	✓	✓	✓	✓
Model Taxonomy	✓	✓	✓	✓	✓	✓	✓
Evaluation & Benchmarking	✓	✓	✓	✓	✓	✓	✓

This review integrates scattered research on EEG emotion recognition into a cohesive view. It provides an overview of the essential challenges of EEG signal processing, feature extraction, and deep learning for modeling emotions and comparisons of popular neural architectures like convolutional networks, recurrent networks, and graph networks for subject-specific and subject-independent tasks. This review provides an insight into the essential trade-offs of generalization capability, robustness, and complexity that limit the application of the current solutions. Apart from this, the potential of transfer learning and domain adaptation in handling intersubject and cross-domain variability has been discussed with both limitations and opportunities. There are several other topics that will be discussed in the review to introduce an overview of the state of the art while identifying the essential research paths for improving robust and scalable solutions for EEG emotion recognition.

2 Theoretical and Methodological Foundations

The theoretical background defines EEG-based emotion recognition as a framework that links measurable neurophysiological activity to computational representations of affect. It models emotional state as a structured pattern distributed across time, frequency bands, and scalp locations, and it treats learning as the problem of extracting robust features or representations from noisy, non-stationary brain signals [14]. It also establishes the use of emotion label names via two-way paradigms: the discrete classification paradigm, where EEG segments are classified into distinct affective states, and the continuous modeling paradigm, where predictions of valence and arousal ratings are performed [15]. These standpoints shape design decisions, evaluation schemes, and generalizations over subjects and sessions [19].

2.1 Emotion Recognition

Emotion recognition is a principal goal of affective computing in pursuing the aim of enabling computational systems that can efficaciously and adaptively respond to the emotional states of humans. Notwithstanding the increasing penetration of human-computer interaction into more and more areas of everyday life, the overall lack of awareness of user emotional states in human-computer interfaces hampers

the overall efficiency of such interfaces in applications that demand user-level personalization, empathy, and commitment to user well-being, thereby pursuing the need to endow computers with the potential to identify user affective states through some form of user cues or physiological signals.

Conventional emotion recognition methods exploit behavioral features, such as facial expression, speech pattern, body language, and text messages. Though these features express more palatable aspects of human emotion, they are still amenable to control and masking in a controlled manner. In this context, combining different sources of variation with physiological measures, which are less controllable at a voluntary level, constitutes a more objective solution. Among various physiological measures, EEG represents a prominent choice because of its direct sensitivity to neural mechanisms related to processing emotions and its high resolution, enabling the monitoring of rapid dynamics of emotions [15, 20].

EEG-based emotion recognition systems model tasks that relate to EEG-stimuli pairs distributed with certain emotion categories. The changes in the emotional state cause changes in EEG rhythms and spatial activity, as well as changes that could be used to extract new information on state transitions between different emotions. According to a set of recent surveys on EEG-based emotion recognition models, the traditional pipeline of EEG emotion recognition systems includes data processing, noise removal, eigenvector transformation, and machine learning inference, with a current emphasis on end-to-end models. Traditionally, models are based on some hand-engineered features from temporal, frequency, or time-frequency worlds, whereas current models are increasingly based on deep learning models that extract hierarchical features from EEG data.

Deep learning models have characterized the emotion recognition study area because of their ability to learn complex, non-linear correlations in multichannel EEG data. Convolutional neural networks have been extensively employed to leverage the spatial information among EEG electrodes, mostly applied when processing EEG maps or image-like representations of EEG signals. Recurrent and Gated Recurrent Neural Networks have been used to model time information, and their primary purpose is to capture time-dependent information of emotion relevant to the research area. Most of the recent studies incorporated graph neural networks and leveraged the connectivity between EEG channels, which gives neurobiologically valid models for emotion recognition [19, 15].

Despite improvements in algorithm methodology, emotion recognition accuracy is known to remain highly vulnerable to specific experimental conditions. In subject-dependent approaches, the absence of significant variability leads to excellent accuracy, while subject-independent and cross-dataset approaches remain significant challenges [19, 20]. Variability across people and recording devices, as well as divergence across protocols for invoking emotions, causes substantial distribution divergence, thereby negatively impacting overall generalization. Thus, there is a growing trend within recent studies on applying transfer learning, domain adaptation techniques, and alignment techniques for enhanced subject and recording device independence [19, 20].

Emotion recognition is viewed in the literature not only as a classification problem but also as a much more complex inference problem, which is affected by neurophysiological variations, experimental conditions, and generalization. Continued progress in EEG-based emotion recognition, therefore, depends on jointly advancing acquisition protocols, representation learning, and evaluation methodologies to support reliable deployment in real-world affective systems.

2.2 Neurophysiological Foundations of Emotional States

Neurophysiological accounts of emotion in EEG-ER interpret emotional episodes as changes in distributed brain activity that manifest as time-varying and band-limited oscillatory patterns across channels. EEG acquisition during affective stimulation, therefore, produces signals with *spatial structure across electrode locations, temporal structure due to stimulus duration and sliding-window segmentation, and frequency structure because analysis commonly separates δ , θ , α , β , and γ rhythms, each of which exhibits different responsiveness under affective conditions* [14].

Within this framing, EEG-ER pipelines either follow a staged procedure (denoising $\rightarrow$ feature extraction $\rightarrow$ classifier learning) or adopt end-to-end feature learning, and they rely on assumptions about label availability and the degree to which training and test data approximate i.i.d. conditions [19]. When

these assumptions fail—especially under cross-subject, cross-session, or cross-database settings—emotion-related neural patterns appear heterogeneous, motivating modeling strategies that explicitly target domain gaps and subject bias [19]. EEG-ER surveys formalize feature construction as multi-domain representation learning. Temporal features describe transitive or long-term variations of events, while spectral features are the result of band-pass filtering, Fourier transforms, and other transforms like it. Spatial features describe inter-channel relationships that correspond to the spatial organization of the cortical structure of the human brain and its variability [14]. This multi-domain stance links directly to contemporary architectures (CNN/RNN/GNN/Transformer variants) that operationalize spatial-temporal-spectral fusion and to transfer-learning approaches that seek domain-invariant representations under strong inter-individual variability [18].

Benchmark datasets encode these neurophysiological assumptions by pairing affective stimuli (e.g., videos/music) with EEG recordings and either discrete labels or continuous ratings, thereby defining the target emotional state space and the evaluation protocol [19].

2.2.1 Discrete Emotion Representation Models

Discrete EEG-ER models treat emotion as membership in a finite set (e.g., positive/neutral/negative; basic categories), and they implement classification as a mapping from multi-channel EEG representations to class labels. The literature consistently organizes discrete modeling around (a) feature engineering or learned representations across temporal-spectral-spatial domains and (b) classifier families ranging from SVM/KNN to deep neural architectures [14].

A core empirical regularity in discrete EEG-ER is that frequency-domain features—including PSD and differential entropy—serve as widely used affective descriptors, commonly computed per band and per channel before classification [15]. Discrete pipelines, therefore, often couple bandwise features with spatial modeling to reflect the channel topology, using methods such as CSP/HDCA for spatial structure or deep models that internalize spatial priors [15].

Additionally, deep learning reviews break down the most popular discrete EEG-ER models into single models: CNN/RNN/GNN models, attention models, combination models, and domain transfer models or models that perform DG/DPre [18]. According to the categories of the above review, the most popular discrete models are the following: DEAP-discrete models of CNN models [21], hierarchical CNN models of DEAP data that use natural spectral features like differential entropy maps [22]. RNN/GRU/LSTM variants implement the concept of temporal dependency, such as attention-driven LSTM classifiers and spatiotemporal recurrent designs [23]. Parallel convolutional-recurrent designs explicitly fuse spatial filtering with temporal recurrence in multi-channel EEG [24]. GNN-based discrete EEG-ER models represent electrodes as nodes and learn adjacency structures to encode functional coupling patterns [25].

Discrete state modeling also depends on the experimental regime. Surveys separate subject-dependent and subject-independent protocols because cross-subject generalization remains harder and motivates architectural choices (e.g., graph attention, domain adversarial training) and evaluation practices [15]. In this context, domain adaptation methods and adversarial learning explicitly address subject bias by aligning feature distributions across individuals [18].

The transfer-learning approaches for EEG-ER problems further solidify discrete EEG-ER as a problem of learning over gaps within domains owing to differences within the subject, session, or database. Domain transfer for knowledge can be classified along the lines of instance transfer, feature transfer, parameter transfer, and relation transfer, each dealing with the discrete classification problem through minimizing the discrepancy over domains [19]. Methods, for instance, weight assignment as in TrAdaBoost, seek to weight the source instances to align with the target domain [19]. Others apply feature transformation or learn subspaces to reduce distribution discrepancy and include deep learning approaches that learn transfer representations [19]. The methods for parameter transfer encompass model fine-tuning and transductive parameter transfer, making effective use of model portability across domains [19].

Discrete models also incorporate label sets that are specific to the data sets. Discrete sets of categories such as anger, fear, sadness, and happiness, among others, are commonly found in public databases, and they can also help in a comparative study [19].

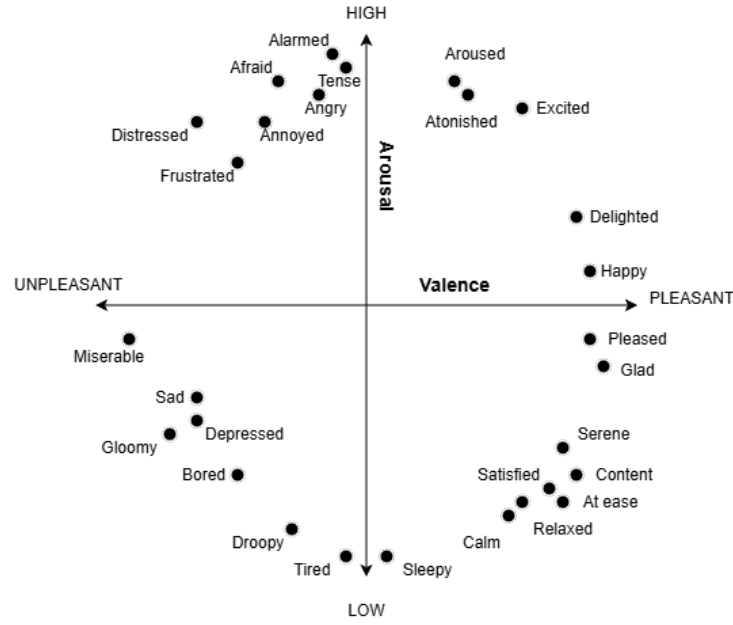


Figure 1: Valence-Arousal 2D Affective Model

2.2.2 Continuous Dimensional Emotion Representation Models

Continuous EEG-ER models treat emotion as a point in a low-dimensional affective space (most commonly valence–arousal, sometimes extended with dominance/liking/familiarity), and they implement learning as regression, ordinal prediction, or discretization of continuous ratings into quadrants/bins. Reviews emphasize that datasets explicitly differ in whether they provide continuous ratings or discrete labels, and they organize benchmark corpora accordingly [14].

From a neurophysiological perspective, continuous modeling aligns with the idea that EEG dynamics vary smoothly over time and that affective response strength differs across frequency bands and windows; thus, temporal segmentation and band decomposition are foundational operations before continuous inference [14]. Continuous systems typically compute bandwise features (PSD/DE/entropy) or learn representations end-to-end, and they map these representations to dimensional ratings using neural regressors or hybrid architectures [19].

Continuous emotion modeling also inherits the central challenge of domain gaps. Transfer-learning surveys define cross-session transfer as addressing acquisition-time variability within the same subject and define cross-subject transfer as addressing individual differences; both scenarios directly affect dimensional prediction because rating scales must remain comparable while EEG distributions shift [19]. Cross-database transfer further compounds these difficulties due to differences in devices, experimental conditions, and label distributions, making it the most challenging scenario for stable continuous inference [19].

In practice, continuous EEG-ER systems often discretize ratings into two-class (high/low) or four-quadrant valence–arousal tasks to enable comparability with classification pipelines while preserving the underlying dimensional semantics [14]. Surveyed results illustrate that some methods explicitly evaluate both binary dimensional splits and quadrant partitions on datasets such as DEAP and DREAMER, reflecting a continuum between regression-style modeling and discretized dimensional classification [14].

Deep learning reviews indicate that architectures used for continuous affect frequently mirror those used in discrete settings—CNNs for local spatiotemporal patterns, RNNs/GRUs for sequential structure, and attention mechanisms for adaptive weighting—while evaluation emphasizes stability and generalization across subjects, which remains more challenging but more deployment-relevant [18]. Another factor, according to survey studies, is that EEG signal processing and feature extraction are still very important, given the presence of noise in EEG signals, which require signal denoising and robust representation learning for predictive processing [19].

Transfer learning becomes especially relevant in continuous affect because it reduces target-domain labeling demand by transferring learned structure from related subjects/sessions/databases, but it also introduces risks of negative transfer when domain gaps are large [19]. Consequently, feature-based deep transfer learning (e.g., discrepancy minimization and adversarial alignment) emerges as a dominant direction for improving dimensional generalization [19].

2.3 Emotion Induction and Stimulus Design

Emotion induction and stimulus design determine how affective responses are reliably evoked and temporally aligned with EEG recordings. Prior methodological surveys describe emotion induction as a prerequisite for meaningful EEG labeling, because weak or inconsistent stimuli directly degrade the validity of learned affective representations [26]. Paradigms involving images, music, or video clips remain popular paradigms because they ensure accurate onset-offset timing, which also ensures easy repeated measurement for every participant, helping in easy segmentation and extraction of features [27]. These paradigms also involve post-trial self-evaluation tasks for mapping brain responses with subjective experience in order to reinforce the accuracy of labeling at the trial level [13].

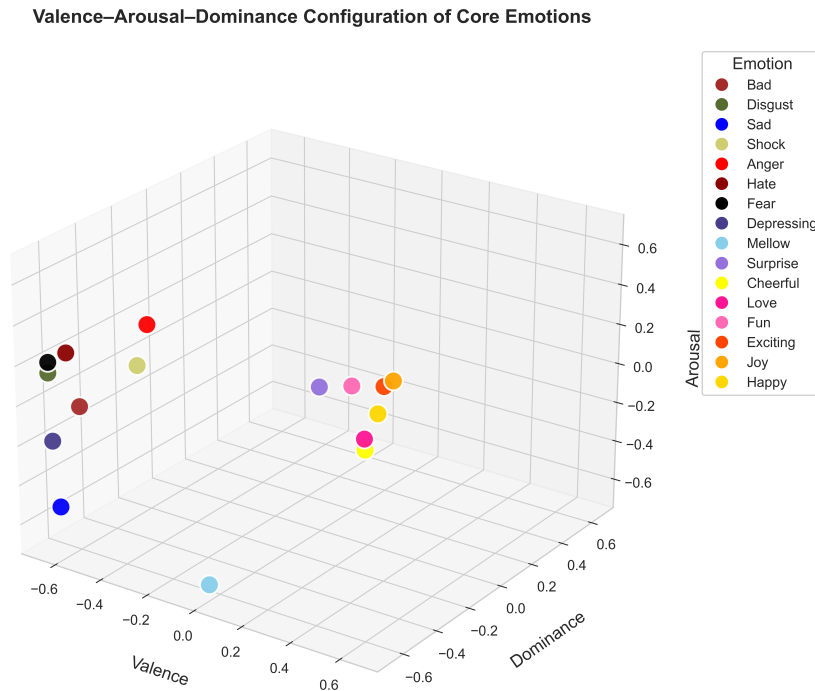


Figure 2: Stimulus

Interactive and task-driven stimuli are growing in usage as alternative paradigms because they promote enduring and contextually rooted emotion processes that passive stimuli may fail to support. However, the literature underscores that such paradigms significantly amplify inter-subject variability as well as the overall complexity of the experiments [28]. Memory recall methods based on the subject's personal memory rely on personal emotion triggers but fall prey to personal cognitive and emotion variables; therefore, their cross-subject universality is limited [29]. In summary, modern studies on emotion processing with the EEG directly address the issue of stimuli as a trade-off between standardization, engagement, and universality [30].

In terms of stimulus modality, the scope of induction practice also extends towards the integration of multimodal signals (such as peripheral physiology and facial expressions) in order to check if the target affective state can be expressed uniformly across the trials, which favors temporal synchronization between subjective states and physiology targets [31]. Cross-corpus evaluation also favors stimulus calibration, as differences in the content of elicitation can contribute towards the shift problem in EEG rep-

representations of affect [32]. In addition, modern learning strategies (e.g., contrastive and self-supervised representation learning) encourage elicitation designs that produce diverse yet label-consistent trials, since representation objectives benefit from repeated exposure to comparable affective episodes under varying contexts [33]. These developments sit naturally within the continuous affect tradition, where valence–arousal modeling formalizes emotion as graded dynamics rather than rigid categories. Finally, the broader affective-computing literature shows that robust elicitation and labeling practices generalize across biosignals (e.g., ECG as well as EEG), reinforcing the view that stimulus design functions as a system-level determinant of downstream generalization [34].

2.4 Progress in Data Acquisition Technologies

The development of EEG acquisition equipment is the basis for the development of EEG-based emotion recognition research. The early study of affective brain activity mainly used research-grade EEG systems characterized by high-density electrode arrays, gel-based sensors, and shielded laboratory settings; these provide a high signal-to-noise ratio and also very accurate spatial resolution, which is a requirement for finding the cortical dynamics of emotions reliably. However, their high cost, time-consuming setup, and participant discomfort make it difficult to scale and be practical for real-world scenarios of emotion recognition. As emotion recognition research gradually shifted from highly controlled laboratory studies toward applied and real-life contexts, the limitations of traditional EEG hardware became obvious.

EEG acquisition hardware also plays a significant role in the data's quality, the model's generalization ability, and the system's feasibility for use in the real world for emotion recognition systems. Hardware factors such as EEG electrodes, channels, stability of signals, and resistance to movements also affect the reliability of the acquired feature for emotions. Emotion recognition systems trained to recognize data from research acquisition systems find it difficult to generalize and recognize signals originating from consumer acquisition systems or wireless acquisition systems for data collection. Thus, acquisition system selection is not just a design choice but now a deliberate factor in model performance between subjects, data set transferability, and feasibility for use in the real world for human-computer interaction, wearable healthcare applications, or affective computing systems.

Hairston et al., [35], in their discussion on various usability problems in conventional EEG methods used for affective computing, focus on the problems caused by the use of 'gel electrodes' and the 'lack of mobility' of the user, though these methods are sure to produce a high degree of data clarity. Debener et al. [36] present the use of mobile EEG equipment that allows for the performance of tests while performing naturalistic procedures, showing the viability of the analysis of related EEG-based emotions. Nevertheless, the raised sensitivity to motion and environmental noise is considerable. On the other hand, wearable equipment with EEG systems, including wireless transfer technology for the continuous monitoring of patients, has also been suggested by Casson et al. in [37]. Although the use of such equipment promotes mobility along with compliance in patients, the number of channels in wearable equipment tends to be reduced, which might influence the emotion classification accuracy. Grozea et al., in their article [38], develop a type of dry electrode based on a bristle to reduce gel preparation without adversely affecting signal quality. This reduces not only the discomfort in gel preparation but also improves overall signal quality through minimal head movements or extended period recording. Huang et al. [39] proposed comb-shaped active dry electrodes for improved scalp coupling. As more robust design solutions are developed for dry electrode upgradation, the increased complexity and cost of such manufacturing processes hinder their acceptability. Zhao et al. [40] explore short-calibration EEG systems with reduced channel configurations in emotion recognition, demonstrating that affective cues can be preserved even through an optimized preprocessing approach. Nevertheless, the reduced spatial resolution limits the analysis of fine-grained patterns of emotions. To compare the performance of real-time EEG signal processing carried out with consumer-grade devices, the paper by Mullen et al. [41] shows the impact of domain shifts due to differences in equipment and the need for effective learning schemes. OpenBCI studies [42] have shown that low-cost and open-sourced EEG data acquisition platforms are well-suited for large-scale experimentation efforts. Nevertheless, low fidelity and improper placement of electrodes are still

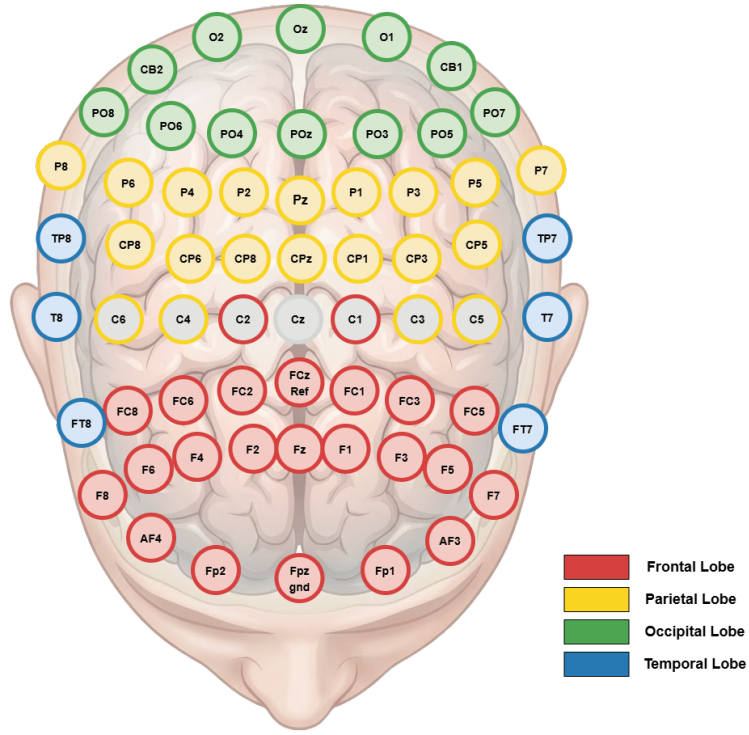


Figure 3: EEG Electrodes

Table 2: Association between major cerebral regions and commonly observed EEG frequency bands.

Brain region	Dominant EEG frequency bands
Frontal lobe	Theta (4-7 Hz), Beta (13-30 Hz), Gamma (>30 Hz)
Temporal lobe	Theta (4-7 Hz), Alpha (8-13 Hz), Gamma (>30 Hz)
Parietal lobe	Alpha (8-13 Hz), Beta (13-30 Hz)
Occipital lobe	Alpha (8-13 Hz; dominant), Beta (13-30 Hz)
Central (sensorimotor cortex)	Mu rhythm (8-13 Hz), Beta (13-30 Hz)
Limbic-related regions (indirect EEG influence)	Theta (4-7 Hz), Gamma (>30 Hz)

some key barriers to achieving the goal of accurate emotional recognition.

The advancement of EEG data acquisition equipment can be seen as a good example of a trade-off balancing the quality of the data with the real-world applicability of the equipment itself. Investigational equipment maintains the ability to acquire the best quality data in terms of foundational research, but wearable-style equipment can greatly help in the scalable use of emotion recognition in real-world applications. Although the advancement in terms of the use of dry sensors, wireless data transfer, and in-data-conditioning can help in the reduction of the barriers associated with the use of the equipment itself, the process still needs the acknowledgment of the importance of the role of EEG data acquisition equipment in the emotion recognition system itself, balancing all the strategies in the process in order to achieve the most successful result in real-world use cases.

2.5 Multimodal Emotion Modeling

Multimodal emotion modeling appears as an alternative solution for the deficiencies in unimodal systems due to their susceptibility to noise and contextual variations in interpreting the emotional state [43]. Nevertheless, human emotional responses arise through various modalities such as facial, vocal, and physical communication, which convey valuable personal affective-related communication in addition to individual sources [44]. Thus, through the process of integrating multiple sources of communication, the

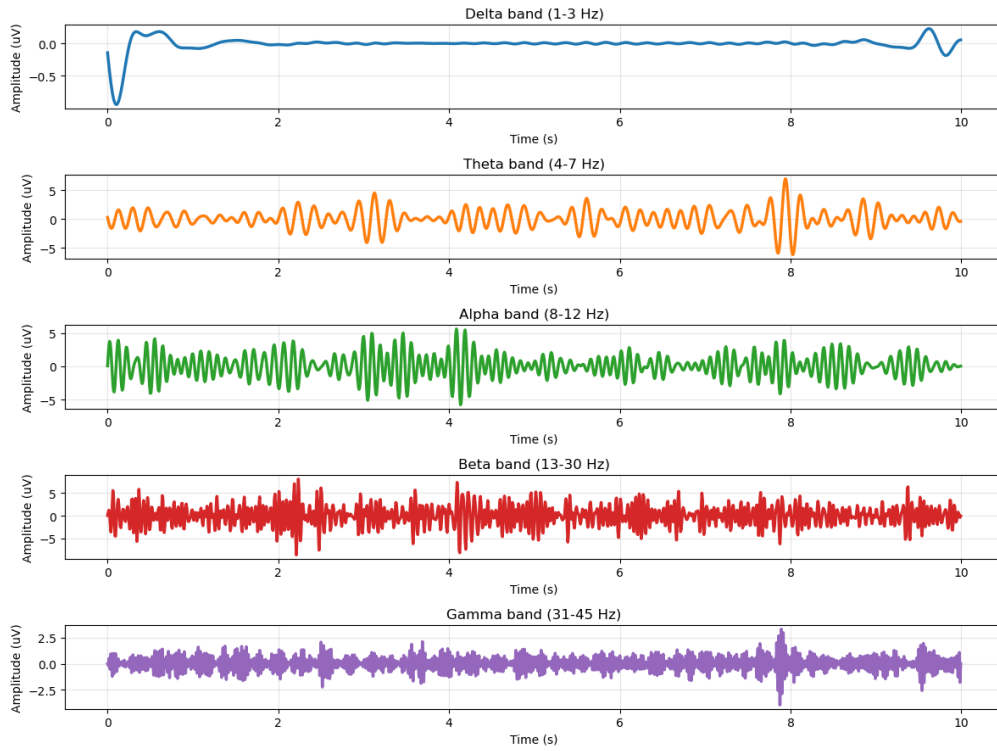


Figure 4: Bands

overall view and understanding of personal emotional state and responses remain contextual [44]. Emotion analysis in a multimodal environment merges behavioral responses such as facial, speech, text, and gesture signals with physiological signals including EEG, ECG, EMG, GSR, and respiration [45]. External behavioral responses mainly focus on actual emotional responses. EEG focuses on internal brain aspects of affective responses that are inaccessible for conscious experience [15]. EEG helps focus these responses in a temporal manner, relating them to actual emotional responses, at the neural level. They play a significant role in responding to subtle affective responses that might not be captured by other signals. The blend of EEG with the above signals helps make these responses more universally applicable.

It is reported in the literature [46] that multimodal fusion paradigms, each of which corresponds to a different level of integration of cross-modal information, such as *early*, *intermediate*, and *late*.

Chen et al. [47] investigate early fusion by combining features from EEG, facial, and speech modalities before classification; improved results are shown when feature normalization and synchronization are appropriately applied. However, early fusion is generally plagued by the growth of dimensionality as well as increased sensitivity to misalignment across modalities, complicating training and potentially degrading performance when features are high-dimensional or poorly synchronized. Kessous et al. [48] have also applied feature-level fusion in multimodal emotion tasks, where the results show that appropriate preprocessing enhances recognition but such approaches remain challenged by noise and feature heterogeneity across channels.

Proposals for intermediate fusion methods have been developed by Shin et al., [49] in which asynchronous modality combination makes use of temporal modeling and sequence alignment. Although this method improves on handling temporal discrepancies between modalities such as EEG and other modalities, it imposes additional complexity in terms of computation. Furthermore, in a different approach based on deep contextual modeling of asynchronous modalities, Liu et al. [50] utilize intermediate fusion; however, this has limitations in terms of high requirements for training data and in handling long-range dependencies.

Wang et al. [51] delve into a decision-level fusion where outputs from independently trained classifiers in EEG, ECG, and GSR are combined by weighting and voting mechanisms; this imposes greater ro-

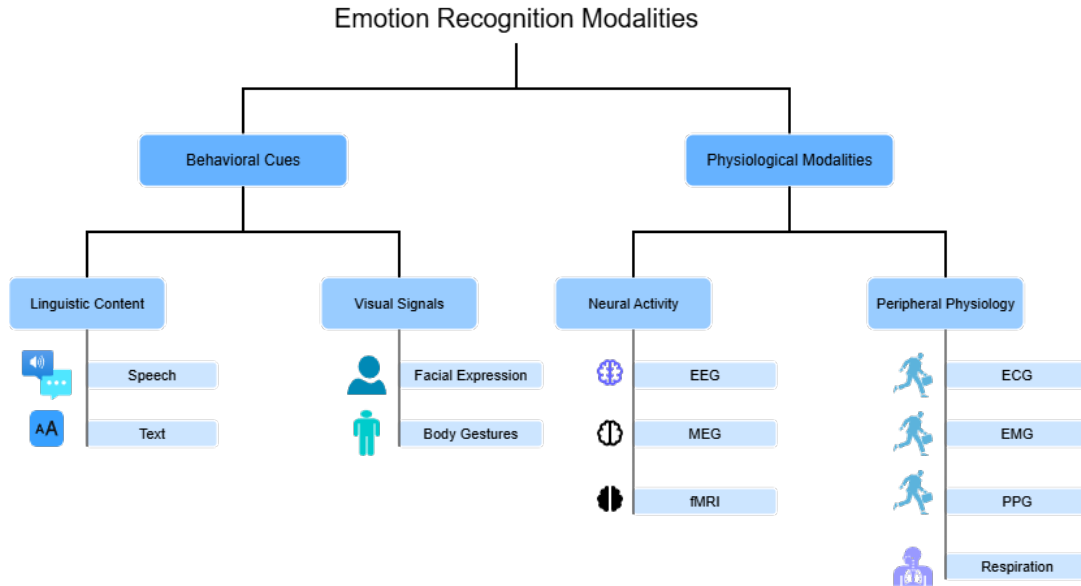


Figure 5: Modalities

bustness and flexibility within the inclusion of modalities. The technique avoids the rigid constraints of synchronization and also allows for the easy integration or exclusion of modalities but does not explore the deep intermodal relationships and may underutilize channel-correlated information. Soleymani et al. [43] review decision-level fusion on physiological systems and present improved generalization and discuss how feature interactions are not fully exploited.

Mittal et al. [52] have developed multimodal deep learning models that utilize attention mechanisms for modality contribution weightings depending upon their importance in a certain context, thus reducing the impact of unreliable modality contributions. Despite these models providing good contributions to this area by reducing the effects of unreliable contributions to a decision that is derived from a multimodal approach, they are still data-intensive optimization models that may not effectively succeed in the absence of available data. Priyasad et al. [53] have taken this concept a step further by developing attention-based models for multimodal fusion that have improved outcomes but are facing issues of model interpretability and increased computation.

The stability of affective patterns in multimodal datasets, such as in DEAP datasets, can improve emotion recognition according to Zheng et al., [54], although certain problems in bias and data modalities are still present. Similarly, Correa et al. [55] employ AMIGOS multimodal data sets and are successful with subject-independent improvements, although large inter-subject variation in multimodal combinations remains a problem in achieving higher results in this area. From a review paper by Ma et al. [15], where certain improvements in the specified area are discussed, it's noted that multimodal data sets are constantly more successful compared to their unimodal analogs and are working toward standardizing certain techniques in data evaluation.

Multimodal emotion modeling has been identified as a development that enhances the process of affective computing by incorporating EEG along with behavioral and physiological modalities to help computers effectively address the challenges that may come about due to the utilization of a single solution to the process. Although various fusion methods have been identified to have different advantages that may come in handy during the modeling process, they present various complexities, synchronicity considerations, or robustness during a fusion evaluation process. Multimodal models have been widely recognized for their incorporation of DL to increase adaptability to different forms of diverse modalities, which may come in handy during the decision process. Even though utilizing DL may demand extensive learning resources, various literature has been published to prove that the utilization of multimodal methods is significantly superior to the utilization of unimodal methods, as confirmed by literature focusing on various multimodal datasets such as DEAP, AMIGOS, MAHNOB-HCI, or MPED datasets.

From the literature, multimodal methods have been recognized as a fundamental necessity for creating an effective EEG-based modeling approach suitable for operation in a realistic human-computer environment.

2.6 EEG Correlates of Emotional States

The essential basis of deriving emotion through an EEG remains rooted in the principle of recognizing the inherent rhythmic activity occurring over the human cortex as an outcome of the processing of emotion. It must also be noted here, unlike the conventional belief, the processing of an emotion is rarely indicated through the peak intensity of the EEG signal, as the expression of emotion largely remains indicated through a combination of dynamic activity occurring over a certain range of frequency bands. These, in turn, manifest as a product of the inherent neural activity underlying the emotion possessed by a particular individual [18].

This view has driven emotion modeling from single-feature representations toward multiband, spatially distributed, and temporally evolving neural descriptions. The emerging evidence suggests that emotionality arises from the interaction of rhythmic modulation and interregional coordination, motivating feature extraction and representation strategies that explicitly capture frequency-specific activity, spatial asymmetries, and functional connectivity.

Understanding the spectral and connectivity correlations for emotion is at the heart of how to develop efficient EEG-based emotion recognition systems. For example, frequency bands such as alpha have proved to have a certain level of consistency and subject stability of the scheme of emotions and therefore perform well to represent states of emotions for different subjects [18]. As a result, alpha bands have been deemed efficient for robust feature selection for subject-dependent and subject-independent emotion recognition systems.

Unlike alpha bands, the involvement of the beta band in emotional states, especially during heightened emotional states, is well evidenced, particularly in prefrontal regions, where a link to cognitive-emotional control parallels the involvement of arousal-related emotional regulation [56]. In addition, delta and theta bands have been found to carry complementary information, with delta band activity related to emotional salience, while theta band connectivity is related to emotional integration/coordination [57]. More importantly, emotional states are also quantified through inter-channel correlations, where larger discriminability is observed, to obtain a holistic picture of neural interactions. Using these subspaces, finer quantification of emotional states is observed based on inter-channel correlation patterns, ensuring that a discriminative emotional state is quantified over localized band power features [58]. The time alignment effect is also observed to have an essential impact on neural components based on the emotional stimuli, reinforcing its importance in maintaining neural state changes in decision-making based on affect labeling [59]. Additionally, when basing research on interpretability, there is a need to quantify emotion based on significant patterns to avoid biases in the data set, leading to the wrong neural components [60].

Meanwhile, a comprehensive collection of research on EEG-based emotion recognition was conducted and summarized by Geng et al. [18], during which the discussion focused on the characterization and importance of alpha activity when recognizing and understanding overall and general emotional dynamics and how sensitivity to emotion can benefit and possibly be enhanced through the exploration and investigation of other frequency bands, apart from the alpha band. A disadvantage and a possible drawback hidden in their discussion is the fact that these findings were summarized at a conceptual level. Sun et al. [56] introduced the dual branch dynamic graph convolutional approach that directly addresses the beta-band activities, as well as the spatial relationships among the electrodes. This model proves that the emotional intensities, including the negative arousal, are indeed significantly detected in the modulation of the prefrontal beta signals. The process remains susceptible to cognition because the beta signals are also impacted by non-emotional cognitive activities. Zhang et al. [57] proposed a multimodal functional connectivity approach that could simultaneously analyze delta, alpha, beta, and theta frequency bands. The results showed that there were significant emotion-related correlations in delta frequency bands and additional discriminative information from alpha and beta frequency bands. In a study

using the correlation-based approach for various frequency bands such as alpha, beta, and gamma frequency bands. The theta frequency band contributes mainly through the connectivity measures. The limitations of the study were the absence of explicit interaction modeling, which limited the value of the approach. Choong et al. [58] have attempted to investigate the use of correlations among alpha, beta, and gamma frequency band connectivity features and how the connectivity features under this use case have accounted for the identification of relevant emotional interactions, even when these interactions do not manifest as identifiable features across individual channels. Liu et al. [59] used movie-induced affective stimuli to retain the temporal traces of neural activities and provide time-aligned labels of affective states. While this method is effective for capturing dynamic responses related to emotions, it unfortunately depends highly on the inter-subject consistency in the perception of emotions, which may not generalize well to a truly diverse population or across different cultural contexts. Qing et al. [60], on the other hand, emphasized the aspect of model interpretability with the proposed mechanisms that connect the decision process with discernible patterns present in the signal. This enables a specific approach to validate that the underlying process utilizes neurophysiologically plausible features. However, the overall level of model interpretability is still hampered by the inherent complexity.

Collectively, this set of research supports the existence of a general neurophysiological framework for emotion states as encoded by both frequency-specific modulation, spatial organization, interregional connectivity, and even dynamics. Emotional states will be represented in the brain as an ensemble process of activity over multiple frequency bands and brain regions rather than as an expression of some global brain state. Clues provided by alpha and beta rhythms will be useful for both stability and intensity information, while salience and interregional integration will be provided by the delta and theta bands, with relational information provided by functional connectivity for differentiation among emotion states. Temporal dynamics and interpretability again increase the relative neurophysiological validity of emotion state representations to support EEG feature and modeling designs that well exceed local amplitude-based emotion recognition methods.

2.7 EEG Signal Processing (Content to be humanized)

The objective of EEG signal processing in emotion recognition is to transform the raw and noisy EEG signals into representations that are invariant to inter-subject and inter-session variability, while capturing the neurophysiological dynamics that are relevant to emotion. In this regard, the signal processing pipeline is equivalent to a series of conditioning steps that include: (i) suppression of non-cerebral noises, (ii) standardization of the temporal structure to facilitate learning, and (iii) definition of feature spaces that reveal the temporal, spectral, and spatial characteristics of emotional states. The signal processing pipeline is not just a pre-processing step, it is a critical component that determines the information that can be extracted and utilized by the models, and it affects the robustness, interpretability, and generalizability of the models directly.

Affective EEG is usually measured using a specific protocol in which the emotional response is dynamic over time. Therefore, the processing pipeline must accommodate the ability to represent transient responses (for example, early perceptual responses) and sustained responses (for example, sustained affective responses). As the datasets move away from the controlled environment, the pipeline focuses more on the resilience of artifacts and stability across domains, encouraging the use of structured denoising, segmentation, and representations that align with the current deep learning encoder models [61].

2.8 Preprocessing

The signals received through an electroencephalogram are known to be quite noisy and prone to a number of artifacts, and therefore preprocessing is an essential step for the recognition of emotions through EEG signals. The preprocessing step is concerned with the cleaning and preprocessing of the raw EEG signals to improve their quality and normalize the input signals for the subsequent steps of feature extraction and classification. The preprocessing of EEG signals is an essential step for improving the SNR of EEG signals and removing external noise and artifacts such as power line noise and physiological ar-

tifacts such as eye blinks and muscle movements for better performance of the classification model for emotion recognition. It is thus important to apply proper filtering, artifact removal, normalization, and other cleaning procedures before the feature extraction step. Moreover, the use of similar preprocessing procedures such as the use of similar filters and scaling helps to eliminate variability and errors in the comparison of signals from different domains. Below, we discuss a variety of critical preprocessing procedures and approaches used in recent studies on EEG-based emotion recognition.

2.8.1 Filtering and Noise Reduction

The process of filtering and noise reduction is one of the fundamental steps in the preprocessing of EEG signals, in which the raw EEG signals are cleaned of various types of non-neural artifacts while keeping the relevant neural components. The EEG signal is very susceptible to external and internal noise, and thus filtering is critical in ensuring that the final signal is indicative of the cognitive or affective processes in the brain instead of other unwanted processes.

The filtering process is of utmost importance in the context of emotion recognition using EEG signals. Emotional states in the brain have very subtle effects in the standard EEG frequency bands, including delta (δ), theta (θ), alpha (α), beta (β), and gamma (γ). However, these effects can very easily get buried under unwanted effects caused by low-frequency drifts (such as skin potentials or sweating), high-frequency muscle movements (EMG), or electrical noise such as power line noise.

The noise reduction techniques that are generally used are band-pass filtering for restricting the noise band to biologically relevant frequency ranges, notch filtering for eliminating noise components present at particular frequencies such as 50 Hz or 60 Hz, and spatial filtering for reducing the effect of noise artifacts. The more complex noise reduction techniques may include complex filtering methods such as regression analysis for reducing noise artifacts and blind source separation for reducing noise components. The type of filtering used, i.e., linear filtering, Butterworth filtering, and Chebyshev filtering, is important for avoiding phase distortion of emotional responses.

Even with these techniques, it is difficult to achieve an optimal SNR. The filtering techniques are such that they can eliminate genuine low or high frequency components of brain activity, which are characteristic of emotions. It is therefore critical to properly calibrate and validate filtering techniques to optimize noise reduction while preserving relevant features of the signal.

2.8.2 Artifact Removal

Although filtering can remove a wide range of spectral artifacts in EEG, it is sometimes inadequate to deal with non-stationary, physiological artifacts that may be generated by eye blinks, eye movements, facial muscles, and jaw activity. These artifacts may introduce sudden and large amplitude artifacts that can severely contaminate the EEG signal, especially in frontal and temporal areas that are very important for affective processing. Hence, an additional step for the removal of such artifacts is necessary to ensure that the brain activity associated with emotions is not masked by the presence of such spurious signals.

In the case of EEG-based emotion recognition, the need for effective artifact removal is even more important because of the fine-grained nature of the emotional state indicators in the neural activity. A single blink or facial movement can introduce amplitude spikes or frequency distortions that can dominate the weaker emotional signals in the data. Moreover, because emotional stimulation often involves visual or auditory stimuli, the presence of these artifacts is more likely during experimental procedures.

The general objective of artifact removal techniques is centered on separating and reducing these non-neural components without eliminating important brain activity. Blind source separation, as well as Independent Component Analysis (ICA), may be used in separating the EEG signals into statistically independent components. Components that may contain artifacts may then be reduced in intensity or removed from the signal. Other approaches may also be employed, like regression-based correction, auxiliary channels like EOG, and machine learning algorithms that may be trained in identifying patterns of artifacts.

While these artifact removal techniques are effective in improving the clarity of the signal, there are certain limitations that may be encountered in using these techniques, such as subjectivity in recognizing certain components of the signal, eliminating important brain activity, as well as the computational costs that may be required in real-time analysis. This has also spurred certain research into developing automated systems that may be able to effectively separate artifacts from important activity, especially in emotion-related analysis where signal clarity is important.

2.8.3 Hybrid ICA-Wavelet Techniques

Although the effectiveness of pure methods such as ICA or wavelet transform may be extremely high for removing certain types of noise from the EEG signals, the use of combinations of such techniques may prove to be even more effective. Recently, the use of hybrid models of ICA and wavelet transform techniques has been proposed as an effective tool for the cleaning of EEG signals. The use of such techniques may prove to be extremely effective for the cleaning of EEG signals, especially in the context of emotion recognition, where the neural patterns are extremely subtle and are highly susceptible to being affected by various types of noise such as those due to eye blinks, muscle movements, etc. The use of such techniques may help in the cleaning of the EEG signals without affecting the emotional content.

In the area of emotion recognition, where the patterns of neural activity associated with emotional states are delicate and vulnerable to distortion by eye blinks, muscle activities, and external noise, these hybrid models can be particularly useful. These models open up the possibility of removing artifacts without damaging the emotional information conveyed by neural activity. As EEG-based affective computing systems become more sensitive and widespread, the need for automatically filtering physiological noise becomes more pressing.

Generally, hybrid ICA-Wavelet approaches begin by carrying out a statistically independent component analysis of the EEG data, which helps in the separation of the neural and non-neural components. Instead of completely removing the independent components identified as noisy, as is the case in ICA, the hybrid ICA-Wavelet approach uses wavelet analysis for the separation and elimination of the noisy components. This ensures that the final clean data include only the required components of the brain activity. This type of analysis, which considers space as well as time, is significant in dealing with the complexity of real-world data.

Although these hybrid ICA-wavelet methods are promising, certain trade-offs are also involved in their implementation, which may limit their performance depending on the type and amount of artifacts in the data, as well as the type of wavelet functions and levels of decomposition that are used in the process. Nevertheless, the impact that these hybrid methods have shown in improving classification performance in EEG emotion recognition systems shows their potential as an advanced preprocessing technique.

2.8.4 Channel Selection and Spatial Filtering

Channel selection and spatial filtering form part of the basic preprocessing mechanisms that focus on improving signal quality by optimizing the spatial representation of the EEG signal. Rather than considering all channels of the signal, these mechanisms focus on the most important parts of the scalp while avoiding the effects of noisy and irrelevant channels. This becomes particularly important when dealing with emotion recognition via EEG signals, where emotion signals tend to be localized and easily obscured by irrelevant neural and physiological activities.

In affective computing, it has been found that the selection of important regions of the brain, particularly those that exhibit emotion, such as the prefrontal and temporal regions, can significantly improve emotion classification. By selecting a specific subset of the electrodes, it becomes possible to reduce dimensionality, eliminate noisy channels, and simplify the process. Spatial filtering improves the clarity of the signal by re-referencing the EEG signal and eliminating global noise components, thus enhancing the signal corresponding to localized neural patterns of emotion.

Several strategies are used in this context. For instance, manual or data-driven channel selection methods are used for prioritizing certain electrodes based on neuroscientific knowledge or feature contribution metrics. This ensures that only discriminative channels are used in the process. Spatial filters like CAR and Laplacian filtering are used for reducing reference bias and eliminating common-mode noise. These filters are used for normalizing the EEG inputs for different channels and for highlighting spatial gradients that are related to emotional processes.

While these strategies are useful in several ways in terms of reducing computational requirements and improving robustness, they are not without their limitations. For instance, too much reduction of channels in this context would lead to the loss of emotional signals that are spatially distributed. Spatial filters would lead to signal topography distortion if not used properly.

Table 3: Representative studies on EEG preprocessing techniques for emotion recognition.

Study	Approach	Preprocessing Method	Advantages	Limitations	Key Findings
Somers <i>et al.</i> [62]	Multi-channel Wiener filtering for artifact suppression	Artifact Removal	Preserves neural signal while reducing ocular/muscle noise	Needs covariance estimates; less effective for nonstationary noise	Improves EEG signal quality before emotion decoding
Kang <i>et al.</i> [63]	ICA with heuristic artifact-aware rejection	Artifact Removal	Effectively isolates and removes major EEG artifacts	Manual component selection may reduce reproducibility	Boosts classification accuracy post-cleaning
Phadikar <i>et al.</i> [64]	Wavelet thresholding for blink artifact removal	Artifact Removal	Fast and adaptive for blink noise	Limited to blink artifacts; not general-purpose	Strengthens EEG robustness in emotion pipelines
Koelstra <i>et al.</i> [65]	Bandpass filtering with baseline normalization	Channel Selection and Spatial Filtering	Reduces noise and inter-subject variability	Requires baseline segment; may remove useful bands	Improves single-trial emotion classification
Mahajan & Morshed [66]	Wavelet-enhanced ICA artifact removal	Hybrid ICA-Wavelet Techniques	Automated ocular artifact suppression	Mainly targets eye-blink artifacts	Enhances EEG quality for emotion analysis
Val-Calvo <i>et al.</i> [67]	Lightweight real-time preprocessing pipeline	Channel Selection and Spatial Filtering	Low-latency and computationally efficient	Simplified features limit generalization	Enables real-time EEG emotion recognition
Dura & Wosiak [68]	Channel selection strategy to reduce electrodes for deep learning emotion ER	Channel Selection and Spatial Filtering	Reduces channels while retaining discriminative information; lowers computation	Strategy may be dataset/model-dependent; risk of losing spatial cues	Shows that careful channel reduction can maintain emotion classification performance
Mammone & Morabito [69]	Enhanced AWICA (wavelet-assisted ICA) for automatic artifact suppression	Hybrid ICA-Wavelet Techniques	Automated removal with reduced manual intervention; effective on nonstationary artifacts	Performance depends on decomposition settings; may distort low-amplitude neural components	Improves EEG cleanliness via wavelet-based component processing before downstream analysis
Kumar <i>et al.</i> [70]	Wavelet-enhanced ICA for ocular artifact removal in EEG	Hybrid ICA-Wavelet Techniques	Better EOG suppression than plain ICA; preserves more neural content using wavelet denoising	Computationally heavier than simple filtering; parameter tuning required	Demonstrates improved ocular artifact attenuation with wavelet+ICA hybrid processing

The preprocessing of EEG data is the critical first stage that makes up the entire frame of reference and determines whether brain patterns that correspond to emotional states can be simultaneously detected. The drive for affective EEG signatures to be consistently identified relies on a number of factors such as; their low amplitude nature; the amount of ocular, muscular, and nonstationary artifacts present; etc., and as a result, preprocessing conditions, which can either improve or diminish classification accuracy, generalizability, and reproducibility.

The studies presented in the table can be grouped into four complementary categories. Artifact removal strategies, including multi-channel Wiener filtering, independent component analysis (ICA) based rejection

tion, and wavelet thresholding have been shown to consistently improve the overall quality of the signal, and also have an impact on downstream recognition, but their applicability involves trade-offs such as the validity of the covariance estimated like Gaussian, the subjectivity associated with selecting the appropriate components, or the location of the artifacts being targeted. Hybrid ICA-wavelet approaches attempt to merge the separation component of ICA with the time-frequency denoising capability of wavelets in order to automate the suppression of ocular artifacts, though the benefits of such techniques may be skewed towards blinks and need to be tested on other sources of artifacts. Channel selection and spatial filtering strategies like using baseline normalized bandpass filtering, help to mitigate noise and inter-subject variability, but rely heavily on the availability of baseline data, and may also attenuate informative bands unintentionally. Lightweight real-time pipelines are emphasized to provide low latency and computationally efficient online affective interfaces but not all of the simplifications used in these pipelines are designed to ensure that robustness will be obtained across subjects.

It can be inferred from the Table 3 that preprocessing must be performed according to deployment constraints: aggressive data cleaning is needed when conducting offline modeling; standardized methods should be used in order to provide comparability between preprocessing results; and streamlined processing must occur when carrying out inference in real time.

2.8.5 Feature Extraction

The process of feature extraction transforms the preprocessed EEG signals into a form where emotion-related features become more easily separable. The most important assumption is that affective neural responses occur collectively in time, frequency, and scalp space. Hence, features relevant to emotions cannot be found in a single domain. Feature extraction techniques can generally be divided into two classes: (i) explicit representations, where features can be derived from signals, and (ii) learned features, where deep models learn features from raw or minimally preprocessed EEG or engineered features. The most important trend in EEG-based emotion recognition is the development of structured, model-friendly tensor representations that keep track of inter-domain dependencies. Thus, rather than obtaining separate features for each channel, as in traditional feature extraction, recent pipelines often compute 2D or 3D objects, such as channel $\times$ time, channel $\times$ frequency, or time $\times$ frequency maps per channel, so that CNNs can take advantage of local features. Another trend is based on 3D feature representations along with fully convolutional models, where temporal, spatial, and spectral information is treated as a unified entity, eliminating the need for inter-domain fusion and enabling end-to-end learning of inter-domain interactions [71].

Sequential feature construction is also common due to the temporal nature of emotional responses. Sequential feature construction is also very common due to the nature of emotional responses. Spatial-temporal features can be constructed by stacking window-level features and using sequence models like bidirectional recurrent units to model temporal dependencies among windows. This approach maintains local temporal information while also modeling long-term affective dynamics, which is important when the emotional episodes develop gradually rather than suddenly [72].

Moreover, transformer-aided feature construction can facilitate cross-domain features, as it does in attention-driven fusion. Architectures based on conformer, which integrates convolution for local pattern detection and transformer attention for global interdependency detection, can provide a useful tool for constructing features as local patterns or global connections without resorting to predefined fusion strategies [73]. Additionally, the parallel use of transformers and 3D convolution is also seen to be following the same principle, where the role of the transformer is to identify interdependencies, while the role of 3D convolution is to identify structurally well-defined spatiotemporal relationships that are associated with Emotion concepts [74].

In the majority of emotion recognition systems based on the EEG, the frequency domain features are used, which are derived from the preprocessed signals. Some of the commonly used features, such as power spectral density (PSD) and differential entropy (DE), are widely used to represent the EEG signals in emotion recognition systems. The power spectral density features represent the power of the signal in standard bands such as theta, alpha, beta, etc., which indicates the amount of rhythmic activity present

in the signal in the respective bands. The concept of differential entropy is related to power spectral density, but when we consider a Gaussian distribution of the EEG signal amplitude in a short period of time, the differential entropy h can be expressed as

$$h = \frac{1}{2} \ln(2\pi e, \sigma^2), \quad (1)$$

where σ^2 is the variance of the signal. Note that variance is proportional to signal power in the window, and thus DE is a log-power feature. In effect, calculating DE on an EEG epoch will give a value that is proportional to the signal power in the epoch and can be calculated on a band-by-band basis so, DE for delta, theta, etc. can be calculated in a manner similar to calculating band power.

The power spectral density (PSD) and differential entropy (DE) are popularly used in EEG-based emotion recognition systems, as they are effective in capturing the characteristics of brain signals related to emotions in the frequency domain. Out of these, DE is regarded as a better alternative to power-based features, owing to its numerical stability and normal distribution. The spectral features, in general, tend to be extracted from various bands of brain signals, including delta, theta, alpha, beta, and gamma bands. These features offer a compact representation of signals, which is of a fixed length and can be readily incorporated with ML. Among the advantages that make PSD or DE features noteworthy is that these features remain unaffected by any timing differences that might be present in a trial. In addition, these features, once appropriately filtered, tend to be somewhat immune to high-frequency noise. Yet, by averaging the power over a time window, these features might overlook significant information regarding the dynamics that occur over a more limited period. Additionally, the stationarity of the signal over a given period might be a concern if the emotional changes do not occur uniformly over the course of the stimulus presentation. Ultimately, with regard to these issues, the parameters for the window size, overlap, and frequency resolution must be adjusted properly. While these measures have limitations, the PSD and the DE remain core components in the study of affective EEGs, often providing a foundation for more intricate models.

In summary, the preprocessing step is an essential constituent of the EEG-based emotion recognition approach, which has a direct bearing on the reliability of the results. As can be observed across the literature, there has been an increasing trend towards the development of sophisticated and automatic preprocessing methodologies. For instance, the initial research studies employed basic filtering and manual removal of artifacts, but the latest research studies, including the ones discussed above, have employed sophisticated methods like ICA cleaning, wavelet denoising, and automatic channel selection to clean the EEG signal. The general idea is that the better the signal quality, the better the model will perform.

Another trend observed across the literature is the standardization of the preprocessing methodologies. For example, the data of different subjects or sessions is often passed through the same filtering and feature extraction processes. As the quantity and quality of EEG emotion data increase, and the need for online processing of the data for real-world applications emerges, the preprocessing techniques are shifting towards the development of real-time capable preprocessing methods, reducing the dependency on manual intervention during the process. In the future, the preprocessing techniques are expected to continue the progression towards the inclusion of newly emerging signal processing techniques for the improvement of the quality of the EEG data, thus providing a solid base for the accurate and robust recognition of emotions using brain signals, as the best classifier can only perform as well as the data it is provided with.

Temporal, Frequency, and Spatial Feature Concepts & Cross-Domain Feature Fusion The nature of electroencephalogram (EEG) data is inherently multi-domain: temporal, frequency, and spatial. Each of these domains represents different aspects of brain activity. The temporal domain represents the evolution of brain activity over time, which can be achieved by dividing the EEG signal into time intervals or by employing time series analysis (using windowing techniques). Features in the frequency domain are achieved by transforming the EEG signal (using Fourier transform) and analyzing the signal's power in specific frequency bands: δ (1-3 Hz), θ (4-7 Hz), α (8-13 Hz), β (14-30 Hz), and γ (30-50 Hz). Note that different frequency bands are associated with different levels of emotional content: previous studies have

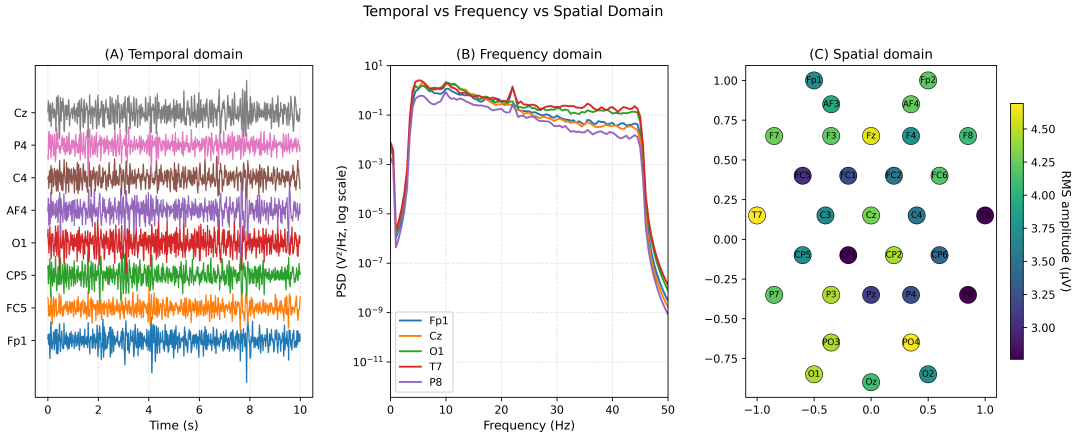


Figure 6: Temporal vs Frequency vs Spatial Domain

shown that specific frequency bands (beta, gamma) are more active during specific emotional content. One of the frequency domain features is differential entropy (DE), which is defined as

$$H(X) = \frac{1}{2} \ln (2\pi e \sigma_X^2) \quad (2)$$

where σ_X^2 is the variance of the signal in the band. Features from multiple bands of the DE provide a compact representation of the EEG signal spectrum, which is useful in emotion recognition. The spatial features aim to capture the distribution of the EEG signal in the brain. For example, it is possible to compute the inter-channel correlation matrix R , which can be defined such that the $(i, j)^{th}$ element is the Pearson correlation r_{ij} between the i^{th} and j^{th} channels,

$$r_{ij} = \frac{\text{Cov}(x_i, x_j)}{\sqrt{\text{Var}(x_i) \text{Var}(x_j)}} \quad (3)$$

which is the adjacency matrix for the graph representation of the spatial features. Thus, the EEG signal provides multi-domain features: the dynamics of the signal over time, the spectral features of the signal, and the spatial distribution of the signal over the cortex. These are called multi-domain features, and the use of all three can provide a holistic view of the emotional state.

Each of these features provides supplementary information for emotion recognition from EEG, and their combination will result in stronger and more accurate emotion recognition models. Temporal features reflect the dynamic nature of emotional responses and their variation. Emotions are processes that depend on time. For example, emotional arousal increases and then decreases over time. Frequency features focus on specific frequency bands of neurophysiological responses that are related to emotional processes. For example, alpha or beta band power increases reflect relaxation or higher engagement. Spatial features reflect emotional processes in specific brain regions. For example, frontal alpha asymmetry is known to reflect affective valence. This is because these three aspects are complementary, and hence their combination can compensate for the limitations of any of these individual feature types. In fact, empirical studies have demonstrated that models that use combined temporal, spectral, and spatial representations outperform those that use individual types of these features. In particular, spatial features are often combined with temporal or frequency features to provide context about where in the brain particular types of temporal or spectral patterns are present. The combination of multiple types of features from multiple domains, therefore, creates a more comprehensive representation of the EEG signal, which can potentially improve accuracy and robustness of emotion recognition against noise or variation.

Fusion is achieved using various architectural styles. First, *tensor fusion* involves the creation of multi-dimensional input representations (e.g., 3D tensors) that enable an encoder network to learn interaction patterns implicitly. Second, *multi-branch fusion* involves the use of dedicated encoders for each domain (temporal, spectral, and spatial) and subsequent fusion of intermediate representations. Third,

attention-based fusion leverages adaptive weightings that highlight the most relevant domain or channel representations depending on varying noise conditions. The use of a dual-branch dynamic graph convolution and adaptive transformer fusion presents an example of the potential for using spatial reasoning and attention-based fusion mechanisms that leverage inter-channel relationships and remain adaptive in feature weightings [75]. For multi-domain EEG encodings, transformer networks are also developed that take in representations from various feature spaces and learn to fuse these using self-attention [76].

Temporal Domain Features: Temporal domain features play a critical role in emotion recognition systems that use EEG as a medium, as emotions are dynamic experiences that change over time. Emotions are dynamic experiences that change over time and are often composed of a series of neural events that can't be described by time-independent features. Temporal features help the emotion recognition system understand how brain activity is changing over time as a response to an emotional stimulus. These features are critical in handling intersubject variability in the timing of the emotion recognition response, as they help the system concentrate on the timing of the signal rather than relying on synchronous stimuli and labels.

Within the emotion decoding paradigm, temporal modeling has been integrated using a variety of deep learning approaches that have taken advantage of the strengths of sequence learning. As an example, recurrent models such as LSTMs or bidirectional RNNs have been widely used to capture the dependencies between successive segments of the EEG signals, modeling the temporal dynamics of affective states. In other instances, hybrid models have been proposed that take advantage of the strengths of both convolutional and recurrent models to learn the spatial patterns and their temporal dynamics. In addition, hierarchical sequence models have been proposed that learn the signals regionally and then aggregate them globally, thus preserving the spatial patterns while modeling the temporal dynamics. Finally, temporal convolutional models have been proposed that offer an alternative to recurrent models, enabling the modeling of long-range dependencies using hierarchical, dilated convolutional architectures.

Although these methods are useful in terms of the recognition of dynamic patterns of emotion, there are challenges that are associated with these methods. These challenges include difficulties in integrating frequency and spatial information or the use of uniform timing. The selection of appropriate window length and temporal resolution is also important in temporal domain modeling. This is important for the avoidance of overfitting or the loss of relevant information, particularly for the improvement of the performance of emotion recognition, especially for naturalistic or continuous emotional scenarios where there is the presence of individual variations of emotional expression timing.

Frequency Domain Features: The features identified by the frequency domain have proven to be a promising approach to study the rhythmicity of the EEG signals, which are closely related to the emotional processes of the human brain. These features have enabled researchers to study the power distribution of the EEG signals over standard frequency bands such as alpha, beta, theta, or gamma waves, which are known to be influenced by emotional processes. Unlike features in the time domain, features in the frequency domain offer a more consistent representation of rhythmic activity in the brain, which is known to endure over sessions and subjects, especially if the EEG signals are appropriately filtered.

Features in the frequency domain are critically important in emotion recognition models, as these offer a means to explore the power in EEG signals in certain bands that is linked to emotional processes, as supported by neurophysiological studies. For example, increased power in the beta frequency band in frontal areas of the brain is known to be related to increased emotional arousal, while power asymmetries in the alpha frequency band are related to valence changes. These features are not easily influenced by timing jitter or noise in EEG signals, which makes them appropriate for real-world EEG-based emotion recognition models.

Several methodologies have been explored for the effective modeling of frequency domain features. For instance, in some methodologies, the EEG signals were converted into spectrograms using various tools such as the STFT, allowing deep learning models to effectively learn the relative contribution of frequency domain features [77]. In other methodologies, the band power and differential entropy were considered

as feature inputs, allowing the model to effectively capture the spectral content of each channel. More recent methodologies have also been explored, allowing the model to effectively capture the spectral relationship between channels using tools such as the convolutional model and attention mechanism [73]. In addition, multi-domain models have also been explored, allowing the model to integrate frequency domain and temporal domain representations for improved emotion recognition accuracy [78].

However, frequency domain modeling for emotion detection with its emphasis on consistent rhythmic patterns has its own set of limitations. For example, there is a need for balance in terms of spectral resolution and computational efficiency. In addition, the parameters of transformations such as STFT are critical and may affect the overall model. Moreover, frequency domain modeling may not account for other dynamics that are critical in decoding complex emotion responses.

In summary, frequency domain features form an essential pillar in EEG-based emotion recognition systems. They are reliable and highly correlated with other emotion-related features. In addition, frequency domain modeling with other domains forms an essential tool in modeling complex affective states with neurophysiological meaning [78, 73, 77].

Temporal-Frequency Domain Features: Frequency domain features are the core of emotion recognition using EEG, as they are based on the rhythmic patterns of the brain's electrical activity, which are directly related to emotional states. The rhythmic patterns of the electrical signals in certain bands of frequencies, such as the delta band (1-4 Hz), the theta band (4-8 Hz), the alpha band (8-13 Hz), the beta band (13-30 Hz), and the gamma band (30-50+ Hz), are known to relate to various aspects of emotion. For instance, the power of alpha and beta rhythms is known to relate to the level of arousal and the valence of emotion. The asymmetry of alpha power in the frontal region is known to relate to approach/withdrawal aspects of emotion. Frequency-specific modulations are generally consistent across subjects and sessions, and hence are useful indicators of emotion classification.

In practice, the analysis of the electrical signals in the frequency domain is achieved through various techniques such as the Fourier transform, the wavelet transform, and the differential entropy method. The above techniques are used to transform the raw electrical signals to a more compact representation. The compact representation is useful because the electrical signals are more susceptible to noise.

Recently, various modeling techniques have adopted frequency domain features in different ways. In some models, frequency domain feature extraction has been performed independently, and the extracted features have been combined with temporal or spatial features to obtain enhanced emotion-specific information. In other models, convolutional or attention-based encoders have been employed to extract frequency domain features, which have been treated similarly to visual inputs to learn abstract frequency domain representations. In addition, various models have utilized graph-based reasoning to extract frequency domain feature-encoded nodes to identify inter-band relationships and regional patterns.

Even though frequency domain-based models have shown promising performance, various trade-offs have been identified in their performance, including sensitivity to the parameters of the decomposition and windowing techniques employed in the models. In addition, models that have adopted frequency domain-based features may overlook essential temporal patterns, thus failing to accommodate different subjects with varying frequency domain patterns. Nonetheless, frequency domain-based features have shown promising performance when combined with other features to provide robust brain state oscillatory patterns and improve emotion recognition systems.

Integration and Fusion Strategies Across Feature Domains: The integration of temporal, frequency, and spatial features has emerged as a primary concern in EEG-based emotion recognition, owing to the multidimensional nature of emotional brain responses. These features describe emotional responses from different but complementary perspectives, where temporal features describe the dynamics of brain responses, frequency features describe brain responses in terms of oscillatory properties, and spatial features describe brain responses in terms of brain regions. Emotion expression in the brain does not occur in individual channels or frequency bands but occurs through interactions across multiple brain responses. Therefore, brain responses that combine multiple views are more expressive and tend to gener-

alize better across subjects and sessions.

The importance of multi-domain fusion becomes critical when handling noisy or uncertain EEG signals. There are scenarios where one domain may not contribute much or may be noisy, such as noisy temporal responses owing to movement, but brain responses in other domains can serve as a fallback, ensuring that fusion-based methods are more robust, especially when handling real-world or cross-subject scenarios.

However, recent developments in multi-domain fusion architectures have focused on flexibility and adaptability. Many of these methods involve multi-branch architectures in which domain-specific feature extractors process temporal, spectral, and spatial information in parallel, followed by fusion layers that combine the features. Some of these methods involve graph convolutional networks (GCNs) to represent spatial relationships between EEG channels, combined with temporal-frequency encoders. Others involve attention mechanisms to dynamically assign weights to features across domains according to their relevance to the emotional state.

There is also an emerging trend towards deeper architectures that learn their own fusion policies implicitly. These include transformer-based feature encoders with domain-specific attention mechanisms and approaches with masked modeling for cross-domain reconstruction. These are very powerful architectures. However, they tend to be computationally expensive, requiring significant memory and regularization during training to avoid overfitting. Moreover, they also present difficulties with regards to their interpretation, as it is not clear what contribution each of the different domains makes towards the overall emotion prediction.

To recapitulate, the integration of various EEG domains can improve the model's ability to learn discriminative and generalizable representations of emotions. Fusion techniques can provide better modeling of emotional patterns, handling of noise, and stability of cross-subject models. Although the above techniques provide considerable benefits, the trade-off needs to be managed for the handling of computational complexities and the interpretability of the models. It is believed that future research will concentrate on the development of more lightweight, interpretable, and data-efficient fusion techniques that can provide the benefits of multi-domain integration applications.

Table 4: Representative studies on EEG feature extraction techniques for emotion recognition.

Study	Approach	Domain Category	Advantages	Limitations	Key Findings
Li <i>et al.</i> [71]	3D spatial-temporal-spectral features with FCNs	Spatial-Temporal-Spectral	Combines rich domain information in unified volume	Heavier computation than 2D encodings	Improves accuracy across datasets
Lew <i>et al.</i> [72]	Spatiotemporal encoding with Bi-GRU	Spatial-Temporal	Captures evolving emotion responses	Sensitive to segmentation granularity	Better for gradual emotion transitions
Song <i>et al.</i> [73]	Conformer model for EEG sequences	Spatial-Temporal	Fuses local and global patterns effectively	Can overfit without regularization	Outperforms traditional CNN/RNN models
Jia <i>et al.</i> [79]	3D DenseNet with attention across domains	Spatial-Temporal-Spectral	Learns dynamic relevance between features	Model complexity is high	Salient EEG areas enhance emotion decoding
Sun <i>et al.</i> [74]	Parallel transformer + 3D CNN fusion	Spatial-Temporal	Separates local/global representation branches	Careful tuning required for balance	Shows strong performance on multi-channel inputs
Yudhana <i>et al.</i> [80]	FFT-based spectral features with k NN classifier	Spectral	Simple and computationally efficient	Ignores temporal dynamics; limited scalability	Shows the viability of classic frequency features
Chen <i>et al.</i> [81]	Differential entropy features with LDA projection	Spectral	Compact and discriminative spectral representation	Linear assumptions limit expressiveness	Improves valence and arousal classification

Continued on next

Table 4 continued

Study	Approach	Domain Category	Advantages	Limitations	Key Findings
Bigirimana <i>et al.</i> [82]	Multi-class CSP spatial filtering	Spatial	Enhances emotion-specific spatial patterns	Sensitive to non-stationarity and subject variation	Boosts multi-class EEG emotion accuracy
Lin <i>et al.</i> [83]	Power spectral asymmetry analysis for emotion classification	Frequency	Captures hemispheric imbalance related to valence	Sensitive to noise and baseline drift	Demonstrates frontal alpha asymmetry correlates with emotion valence
Jenke <i>et al.</i> [84]	Comparative evaluation of frequency-domain EEG features	Frequency	Identifies robust spectral descriptors for emotion decoding	Requires feature selection optimization	Highlights differential entropy and PSD as strong frequency predictors
Al-Nafjan <i>et al.</i> [85]	PSD-based spectral feature extraction with SVM	Frequency	Simple and computationally efficient	Limited temporal resolution	Confirms spectral power variations distinguish valence states
Murugappan <i>et al.</i> [86]	Statistical time-domain features for emotion classification	Temporal	Low computational cost; suitable for real-time systems	Limited ability to capture nonstationary dynamics	Shows time-domain variance and mean features discriminate emotions
Gupta <i>et al.</i> [87]	Hjorth parameters and higher-order statistics	Temporal	Reflects signal activity and mobility characteristics	Sensitive to artifacts	Demonstrates temporal descriptors contribute valence-arousal recognition
Petrantonakis & Hadjileontiadis [88]	Higher-order crossings (HOC) for EEG temporal dynamics	Temporal	Captures nonlinear signal characteristics	Requires careful parameter tuning	HOC features enhance multi-class emotion accuracy
Zhao <i>et al.</i> [89]	Short-Time Fourier Transform for emotional EEG analysis	Temporal-Frequency	Preserves both temporal and spectral information	Window-size selection impacts performance	STFT-based representations improve arousal classification
Subasi & Gursoy [90]	Wavelet transform for EEG emotional state detection	Temporal-Frequency	Handles non-stationary EEG signals effectively	Higher computational complexity	Wavelet coefficients improve emotional discrimination
Huang <i>et al.</i> [91]	Hilbert-Huang transform for nonlinear EEG analysis	Temporal-Frequency	Adaptive decomposition for nonlinear signals	Mode mixing may occur	Enhances representation of emotion-related oscillatory patterns
Li <i>et al.</i> [92]	Feature-level fusion of spatial and spectral EEG descriptors	Fusion	Integrates complementary information	Risk of high dimensionality	Fusion improves robustness across subjects
Zheng & Lu [93]	Combined frequency and spatial asymmetry features	Fusion	Exploits hemispheric asymmetry and band power	Requires careful channel selection	Identifies critical bands and regions for emotion decoding
Zhang <i>et al.</i> [94]	Multi-feature fusion with deep neural networks	Fusion	Learns hierarchical representation from heterogeneous features	High model complexity	Feature fusion enhances cross-subject generalization

The table 4 presents an overview of the various EEG feature extraction methods that have been used for recognizing emotional states. In particular, it categorizes the different feature extraction methods or paradigms according to the type of signals used to construct them: either based on the spectral or spatial domains. The predominant early and light feature extraction methods were primarily based on spectral representation of the signals through the use of the Fourier transform or differential entropy. On the other hand, the predominant features extraction paradigms in the spatial domain were based on filtering techniques that utilized common spatial patterns as the basis for classification.

There is increasing use of these hybrid spatial-temporal-spectral representations in current research, due to their ability to accurately capture the complex and nonlinear characteristics of emotions associated with brain activity. Methods that use three-dimensional feature volume representations, attention-based

methods, and deep learning architectures (convolutional or transformer-based) all combine multi-modal data across domains to produce better, stronger and more precise emotion decoding results. Spatiotemporal sequence modeling methods, such as Bi-GRU and conformer based models use time to encode how emotions are thought to change continuously over time while maintaining the spatial aspect, and have been shown to be very effective for this purpose. The trade-off for these benefits, however, is typically an increase in computational costs and potential for overfitting.

Overall, the progression represented in Table 4 demonstrates a distinct shift towards unified, or so-called multi-domain, deep representations, further emphasizing the significance of so-called joint spatial, temporal, spectral features in modern emotion recognition paradigms based on EEG technology.

2.9 Feature Selection

In emotion recognition datasets using electroencephalogram (EEG) signals, datasets contain high-dimensional feature vectors, including a high number of channels as well as different frequency band features. At the same time, several datasets contain a relatively low number of instances compared to the high dimensionality of the dataset. As a result, there is a great tendency for the occurrence of the curse of dimensionality, which may cause models to overfit as well as possibly ignore several relevant features in the dataset, which often reduces model efficiency as well as generalization of the classifier. Feature selection is one of the main concepts in pattern recognition as well as machine learning that helps reduce a high number of features in a dataset, containing only the most efficient as well as unique features, which can help in improving the efficiency of a classifier in pattern recognition as well as in improving the efficiency of the machines used [20].

2.9.1 Filter Methods

Filter-based feature selection is one of the most used feature selection techniques in EEG-based emotion recognition systems. This is mainly because of its relative ease of implementation, speed, and classifier-agnostic nature. Filter-based feature selection approaches attempt to measure the significance of the features before actually training a classifier. Hence, these are generally employed as a pre-processing step of the machine learning pipeline to reduce dimensionality and improve the learning efficiency of downstream classifiers.

The relevance of the filter method is determined by the nature of EEG data, which is usually characterized as being both high-dimensional and noisy in nature. There is usually an involvement of emotion-related features as well as other features in a pool of redundant and irrelevant information. Filtering out these less informative features at an early stage enables filter approaches to build better classification performance, reduce overfitting, and improve interpretability. This is particularly significant for EEG-based emotion recognition tasks.

There are many different filter-based methods for selecting good features. Two examples of filter selection methods are correlation-based selection and mutual information (MI) selection. A correlation-based method examines the correlation between each feature and the emotional label it is related to in order to find all of the features that have a high correlation with the individual emotional labels and the overall target emotion label. In an MI-based method, the MI score of each feature is calculated with respect to the associated target label as well as with respect to all other features, to assess the relevance and redundancy of each feature. Similarly, each feature in the mRMR framework is evaluated based on its MI to the target label as well as its MI compared to all other features. The purpose of the mRMR framework is to identify a small, non-redundant collection of features that provide the most information when identifying emotion labels. Extensions of the mRMR framework have been developed that incorporate methods such as sequential forward search or floating search, allowing for a larger, more exhaustive search of the feature space with improved efficiency.

Filter-based methods have certain inherent limitations; they may produce suboptimal subsets of features when there are emotional patterns throughout multiple feature combinations or only when comparing each feature separately and independently. Therefore, the subset of features determined by filter meth-

ods will not always correspond to an optimal decision boundary that a classifier might find in an EEG-based emotion recognition system, and so may lead to poor classification performance when classifiers use the features selected from the filter methods. However, filter methods remain viable techniques for EEG emotion recognition systems because of their large-scale ability to apply to general tasks and their ability to be generalized.

2.9.2 Wrapper Methods

Use of wrapper-based feature selection provides an effective method for determining which EEG features are the most informative for classifying emotions; however, the wrapper method is generally more computationally expensive than the filter method. Filter methods assess individual features based on their characteristics or quality, whereas wrappers utilize a learning algorithm when performing the selection process. Therefore, wrappers can assess how well various combinations of features perform during the actual classification process, providing a more tailored and often higher-performing set of features. The utility of the wrapper method within the context of EEG emotion recognition is due in part to the fact that EEG features exhibit complex interactions and nonlinear relationships between them, which is especially prevalent in affective EEG data. Affective EEG data representations tend to display and reflect subtle emotional responses, which may occur simultaneously/ly across many different brain regions and frequency bands, making it necessary to identify interdependencies between EEG features. Given that the wrapper method assesses combinations of the EEG feature subset's performance as a whole, it allows the evaluation and optimization of all combinations of the EEG feature subset for performance on a specified classifier.

Multiple methods showcase the wrapper method. An established technique, Recursive Feature Elimination (RFE), works by sequentially eliminating some of the least effective features, based on how much they influence the classifier's weight or accuracy, thereby refining the size of the feature subset over a series of iterations. Another increasingly popular evolutionary technique, Genetic and Differential Evolution, is used to systematically explore the very large EEG feature space effectively. The feature candidates are continuously evolved from iteration to iteration by simulating biological functions, such as mutation, selection, and cross-over; the evolution process is monitored using the classifier's ability to validate each of those combinations for maximum performance. These methods ultimately produce very effective combinations of features that will contribute to the increased performance and generalizability of emotion recognition across different subjects.

Despite these advantages, the wrapper methods also have their challenges. Their requirement to train the classifiers multiple times as different feature subsets are examined makes the process computationally expensive; this is particularly true when the feature set is large and/or when a computationally intensive classifier is selected for training on the feature set. As such, wrapper methods may not be very useful for practical real-time or resource-constrained scenarios. Another challenge is the possibility of overfitting due to inadequate regularization of the evaluation metric for selecting subsets of features, or when the feature selection method is too closely bound to the specific dataset used for its selection. Nevertheless, wrapper methods continue to remain viable methods for identifying high-quality subsets of features that will enhance emotion recognition for EEG applications.

Table 5: Representative studies on EEG feature selection techniques for emotion recognition.

Study	Approach	Feature Selection Method	Advantages	Limitations	Key Findings
Xu <i>et al.</i> [95]	Orthogonal regression for pairwise redundancy reduction	Embedded	Removes overlapping features, boosts relevance	Assumes linearity in correlations	Improves generalization across individuals
Kannadasan <i>et al.</i> [96]	Differential evolution for feature subset selection	Wrapper	Searches compact, accurate feature sets	Evolution search is slow at scale	Increases classification stability in DEAP

Continued on next

Table 5 continued

Study	Approach	Feature Selection Method	Advantages	Limitations	Key Findings
Liu <i>et al.</i> [97]	EMD with optimal subset evaluation	Wrapper	Efficient combination of decomposition and selection	EMD results vary with input noise	Smaller feature sets yield higher accuracy
Wang <i>et al.</i> [98]	NMI-based EEG channel selection	Filter	Reduces redundancy and sensor count	Subset may vary across subjects	Preserves accuracy with fewer channels
Peng <i>et al.</i> [99]	Minimum Redundancy Maximum Relevance (mRMR) for EEG feature ranking	Filter	Selects maximally relevant yet minimally redundant features	Requires discretization for continuous signals	Improves robustness of EEG-based classifiers using mutual information criteria
Al-Nafjan <i>et al.</i> [100]	ReliefF-based statistical feature ranking	Filter	Fast computation and noise-tolerant ranking	May overlook nonlinear inter-feature interactions	Demonstrates effective dimensionality reduction for valence classification
Jenke <i>et al.</i> [84]	Comparative evaluation of filter-based feature ranking methods	Filter	Identifies most discriminative EEG descriptors	Dependent on classifier compatibility	Highlights mutual information and Fisher score as effective EEG feature selectors
Subasi [101]	Genetic algorithm for optimal EEG feature subset	Wrapper	Explores global search space effectively	Computationally expensive for large feature pools	GA-based selection improves classification accuracy over manual selection
Guler & Ubeyli [102]	Adaptive neuro-fuzzy inference with feature subset search	Wrapper	Combines nonlinear modeling with subset optimization	Risk of overfitting with small datasets	Wrapper optimization enhances emotion-related EEG separability
Shen <i>et al.</i> [103]	Sequential forward selection with SVM evaluation	Wrapper	Produces compact feature subsets	Greedy search may miss global optimum	Reduces dimensionality while preserving classification performance
Guyon <i>et al.</i> [104]	Recursive Feature Elimination (RFE) with SVM	Embedded	Integrates selection into classifier training	Sensitive to parameter settings	Embedded ranking improves generalization in high-dimensional data
Zhang <i>et al.</i> [105]	LASSO-based sparse feature learning	Embedded	Automatically enforces sparsity during training	Linear constraint limits nonlinear capture	Sparse modeling enhances cross-subject stability
Li <i>et al.</i> [106]	Deep neural network with attention-based embedded selection	Embedded	Learns task-relevant features automatically	Requires large training data	Embedded attention improves discriminative EEG feature learning

Table 5 summarizes representative feature selection strategies employed in EEG-based emotion recognition and highlights the evolution from conventional statistical ranking methods to model-integrated sparse learning approaches. Overall, the studies demonstrate that feature selection plays a crucial role in addressing the high dimensionality, redundancy, and subject variability inherent in EEG signals. The table shows that filter methods provide computationally efficient redundancy reduction and channel optimization, making them suitable for real-time or resource-constrained systems. Wrapper methods, although computationally heavier, generally achieve improved classification stability by directly optimizing feature subsets with respect to classifier performance. Embedded approaches integrate feature selection within the learning process itself, enabling simultaneous model training and dimensionality reduction, which is particularly beneficial for deep learning-based frameworks.

Across all categories, a consistent finding is that reducing irrelevant or overlapping EEG features enhances generalization, improves robustness across subjects, and lowers computational burden without sacrificing accuracy. The table therefore reflects a broader research trend toward compact, discriminative, and task-adaptive feature representations, which are essential for scalable and cross-subject emotion recognition systems.

2.10 Challenges and Strategies in EEG Data Handling

Single-modal EEG emotion recognition remains constrained by practical data-handling issues that directly influence generalization, reproducibility, and deployment. Two recurring bottlenecks are (i) redundant or weakly informative feature sets that inflate model capacity requirements and (ii) limited labeled samples that encourage overfitting and unstable training. Recent work, therefore, treats data handling as a co-design problem, where feature selection, augmentation, and generation are integrated into the learning pipeline rather than being optional post-processing steps.

2.10.1 Feature Redundancy

Feature redundancy arises when handcrafted descriptors (or even learned embeddings) contain overlapping information across channels, bands, and time windows, so that many dimensions contribute little beyond correlated noise. Redundancy is especially common when multiple feature families are stacked (e.g., entropy- and spectrum-derived statistics computed per channel) and when spatial neighborhoods yield near-duplicate summaries. A common mitigation strategy is explicit feature selection using filter, wrapper, and embedded schemes, which retains a compact subset while controlling inter-feature dependence.

Redundancy-aware selection increasingly moves beyond simple ranking toward objective functions that penalize pairwise overlap. For example, orthogonal-regression-based formulations incorporate redundancy into the optimization target to suppress mutually dependent predictors while maintaining discriminative utility. A representative direction improves flexible sparse orthogonal regression by adding pairwise redundancy terms into the loss so that the selected subset is simultaneously compact and decorrelated [95]. Complementary work frames selection as a population-based search problem, where differential evolution explores feature subsets under cross-subject constraints to reduce duplication while preserving affect-relevant variability [96]. More recent orthogonal-regression pipelines further combine regression and feature weighting, encouraging sparse, high-utility feature sets that reduce the effective hypothesis space in small-data settings [107].

2.10.2 Limited Data Challenges

EEG emotion datasets typically remain modest in subject count, session diversity, and label density, and labels are expensive because they require carefully controlled elicitation protocols and reliable affect annotation. When deep models are trained on such limited samples, they often latch onto subject-specific idiosyncrasies or session artifacts, which weakens cross-subject portability. Practical pipelines therefore adopt two complementary strategies: (i) augmenting available samples through label-preserving transformations and (ii) generating additional samples via learned generative models.

Augmenting Data Samples Data augmentation creates new training instances by transforming existing trials while maintaining their emotion labels. For EEG, effective augmentation typically respects signal structure (temporal continuity, channel topology, and latent source composition) rather than applying generic image-style perturbations.

Decomposition-driven augmentation is one established approach. Independent Component Analysis (ICA) can separate latent sources, enabling augmentation in component space and reconstruction back into sensor space; this allows recombination-based operations that preserve physiologically plausible structure while increasing trial diversity [63]. Likewise, the Empirical Mode Decomposition (EMD) technique may, through the process, derive the Intrinsic Mode Functions (IMFs), for which the selection process that preserves the high correlation with the waveforms may lead to the new samples [108].

Augmentation also supports semi-supervised regimes where labeled data are scarce. A typical pattern injects weak/strong noise perturbations and uses interpolation between labeled and unlabeled examples to create mixed samples, then aligns the two domains (labeled vs. unlabeled) through domain-adversarial

learning. Parameter-efficient domain-adaptation formulations operationalize this idea by treating interpolation as both augmentation and regularization, improving learning stability under low-label settings [109].

Generating Synthetic Data Unlike augmentation, data generation aims to produce novel samples from learned distributions. Generative adversarial networks are widely in use today, and Wasserstein variants are favored due to their generally more stable training and improved mode coverage. GANs, specifically Conditional Wasserstein GANs, are able to generate diverse feature representations in EEG features conditioned on class information. This serves to increase within-class variability based on the generated samples from classes of interest without having to create direct transformations of the trials [110]. In multi-modal contexts, GAN-based augmentation can generate paired modalities, for example, jointly generating EEG along with another type of signal. This improves consistency across modalities when downstream fusion is used [111].

More recent approaches extend beyond synthesis to a focus on generation through reconstruction, where a model hides parts of the real EEG and tries to regenerate them. This “mask-and-regenerate” strategy produces realistic variants anchored to true recordings, which can be especially helpful when fully synthetic waveforms risk deviating from physiological constraints [112]. Finally, cross-category generation explicitly targets decision-boundary enrichment by creating samples that increase inter-class diversity and boundary coverage; diversity-aware and boundary-aware losses are introduced to avoid trivial interpolations and improve the usefulness of generated data for discriminative learning [113].

3 Modeling and Evaluation Frameworks (Content yet to be humanized)

EEG-based emotion recognition methods structure a classification pipeline that transforms multichannel brain dynamics into affect labels through preprocessing, representation learning, and decision modeling [61]. Contemporary reviews emphasize that deep learning methods increasingly replace handcrafted feature stacks by learning discriminative spatial-temporal abstractions from minimally processed EEG while still requiring careful protocol design for valid performance claims [7, 61]. Model families differ in inductive bias, because convolutional encoders prioritize local spatial-spectral regularities [7], recurrent encoders prioritize sequential dependencies [8], and graph formulations prioritize sensor topology and inter-channel relations [9]. Evaluation settings, therefore, define methodological meaning, since subject-dependent protocols favor personalization, whereas subject-independent protocols favor invariance and generalization across individuals [4].

3.1 Subject-Dependent Analysis

Subject-dependent analysis evaluates EEG emotion recognition models when both training and testing data come from the *same individual*. In practice, for each subject u , the data are split into within-subject train/validation/test folds (e.g., k -fold or trial-wise cross-validation), and a model is trained to capture that subject’s idiosyncratic spatio-temporal and spectral EEG patterns. This setting is widely reported because it often yields higher accuracy than subject-independent testing, providing an empirical *upper bound* on performance when per-user calibration is allowed. However, it can overestimate real-world generalization if the split allows data leakage (e.g., mixing highly correlated segments from the same trial across train/test). Recent surveys and systematic reviews emphasize that subject-dependent results remain common in the literature, but they must be interpreted alongside the chosen splitting protocol and the calibration burden required per user [114, 115, 18].

Song *et al.* [116] did subject-dependent EEG emotion recognition by modeling multichannel EEG as a *dynamical graph* and learning the inter-electrode adjacency matrix end-to-end (DGCNN), with the result being **90.4%** average subject-dependent accuracy on SEED (and **79.95%** under subject-independent cross-validation); limitation: the sizable drop highlights persistent generalization gaps and sensitivity to cross-subject shift. Ma *et al.* [117] did subject-dependent affect inference by a *multimodal residual LSTM*

with weight sharing across modalities, with the result being **92.87%** accuracy for arousal and **92.30%** for valence on DEAP; limitation: the method is multimodal and thus may not transfer directly to EEG-only deployments and can increase sensor/setup complexity. Ding *et al.* [118] did subject-dependent emotion classification by *multi-scale temporal kernels* and *hemispheric-asymmetry-aware* spatial learning (TSception) under *trial-wise* evaluation to reduce leakage, with the result being **63.75%** arousal / **62.27%** valence accuracy under leave-one-trial-out on DEAP (and **61.57%** arousal / **59.14%** valence under trial-wise 10-fold); limitation: accuracy is notably lower than many “segment-shuffled” reports, illustrating how stricter splits reveal the true difficulty and motivating careful protocol reporting. Guo *et al.* [119] did subject-dependent recognition by combining *depthwise convolution* with *Transformer encoders* (DCoT) and channel-importance visualization, with the result being **93.83%** average subject-dependent accuracy for three-class SEED classification; limitation: subject-independent performance remains substantially lower (reported as **83.03%**), indicating reduced robustness without adaptation. Liu *et al.* [120] did subject-dependent recognition with a hybrid *CNN+Transformer* architecture emphasizing interpretability (ERTNet), with the result being **74.23%±2.59** accuracy and **AUC=0.93** on DEAP and **67.17%±1.70** with **AUC=0.91** on SEED-V; limitation: although interpretable, accuracy remains moderate and drops sharply under LOSOCV, indicating that interpretability does not automatically guarantee cross-subject robustness. Zhu *et al.* [121] did subject-dependent recognition by integrating a *self-organized channel graph per band*, pseudo-3D spatiotemporal convolution, attention, and LSTM (SOGPCN), with the result being **95.26%** subject-dependent accuracy on SEED; limitation: the method relies on engineered band-wise inputs and graph construction choices that may not transfer cleanly across datasets/montages. Khan *et al.* [122] did subject-dependent recognition by mapping EEG into *STFT/Mel spectrograms* and fine-tuning pretrained CNNs (GoogLeNet/ResNet) as well as a compact EEG-ConvNet, with the result being up to **99.97%** (STFT-GoogLeNet) and **99.03%** (EEG-ConvNet) on SEED plus a reported **6.5 ms** prediction time; limitation: such extremely high subject-dependent accuracy can indicate an easier within-dataset setting and must be validated against stricter trial-wise splitting and cross-subject/cross-session protocols to rule out optimistic leakage.

$$\min_{\theta^{(u)}} \mathcal{L}_{\text{SD}}^{(u)} = \frac{1}{|\mathcal{D}_{\text{tr}}^{(u)}|} \sum_{(x,y) \in \mathcal{D}_{\text{tr}}^{(u)}} \ell(f_{\theta^{(u)}}(x), y) + \gamma \Omega(\theta^{(u)}), \quad (4)$$

where $\mathcal{D}_{\text{tr}}^{(u)}$ is the within-subject training subset for subject u , $f_{\theta^{(u)}}$ is a subject-specific model (or a shared model fine-tuned for u), $\ell(\cdot)$ is a classification loss (commonly cross-entropy), $\Omega(\cdot)$ is a regularizer (e.g., $\|\theta\|_2^2$ or dropout-induced implicit regularization), and γ controls the regularization strength. In subject-dependent EEG emotion recognition, performance is strongly coupled to *how* $\mathcal{D}^{(u)}$ is split (trial-wise vs. segment-wise shuffling), because temporally adjacent EEG segments within the same trial can be highly correlated [118].

Across recent subject-dependent studies, the best-performing methods under *lenient within-subject settings* are typically spectrogram+CNN pipelines (e.g., EEG-ConvNet and fine-tuned GoogLeNet/ResNet), which report near-saturated SEED accuracy but may be most sensitive to leakage and dataset specificity [122]. Under *more protocol-aware* within-subject evaluation (trial-wise or leave-one-trial-out), TSception provides a clearer picture of difficulty and robustness to leakage, albeit at lower accuracy, making it preferable when methodological rigor and reproducibility are prioritized [118]. For *interpretability with competitive subject-dependent performance*, ERTNet offers strong AUC reporting and interpretable components but still faces major subject-independent degradation [120]. For *capturing inter-channel topology*, graph-based models (DGCNN, SOGPCN) are attractive and can yield high within-dataset accuracy, with SOGPCN reporting a small SD–SI gap on SEED; however, portability across datasets/montages remains a practical concern [116, 121]. Overall, subject-dependent analysis remains valuable as an upper-bound benchmark, but the most defensible conclusions come from studies that (i) use leakage-resistant splitting and (ii) report subject-independent/cross-session results alongside subject-dependent performance [115, 114].

3.2 Subject-Independent Analysis

Subject-independent analysis trains on a source cohort and evaluates on unseen subjects, so domain shift across individuals becomes the dominant error source [123]. Transfer learning and cross-subject adaptation methods reduce this mismatch by learning subject-invariant features through feature alignment, parameter sharing, and regularization, while preserving emotion-relevant variation [10, 123]. Plug-and-play and adversarial domain adaptation frameworks explicitly align latent distributions between source and target subjects using unlabeled target data, so these methods target invariance without requiring full target supervision [124]. Domain-adaptive neural formulations further refine alignment to preserve discriminability under shift [125]. Latent-representation similarity objectives also constrain transfer by regularizing feature geometry between domains [126]. Contrastive objectives encourage subject-invariant embeddings by pulling together emotion-consistent samples while pushing apart emotion-inconsistent samples across subjects [33], and cross-corpus evaluation settings reveal how these invariances behave under dataset-level mismatch [32].

More recent subject-independent methods integrate topology-aware backbones with adaptation modules that manage both cross-subject and cross-dataset variability [127]. Domain-adversarial graph attention models align relational representations across domains while maintaining spatial-temporal structure [127, 128]. Graph attention-based spatial-temporal learning further supports subject-independent decoding by encoding dynamic interactions among electrodes [129]. Multi-source adaptation methods treat each subject or dataset as a distinct source and learn transferable prototypes or shared subspaces [130]. Pairwise and prototypical learning frameworks mitigate negative transfer under heterogeneous sources [131]. Parameter-efficient adaptation formulations reduce adaptation overhead while preserving transfer performance [109]. Broader domain generalization theory motivates training objectives that optimize shift-robust representations rather than single-domain accuracy [132].

Domain shift is a primary reason why EEG emotion recognition models that perform well in subject-dependent settings may fail under subject-independent evaluation. In affective EEG, a domain typically corresponds to a *subject*, *session/day*, or *dataset/device montage*. Because EEG is nonstationary and highly individualized, both marginal feature distributions and class-conditional structures can drift across domains. Consequently, models trained only by empirical risk minimization on multi-subject source data can degrade substantially under leave-one-subject-out (LOSO), cross-session, or cross-dataset testing. A dominant family of solutions is **transfer learning**, which reuses representations learned from source domains and adapts them to target domains via alignment, adversarial invariance, prototype structure preservation, or target-only (source-free) updates [30, 29, 19, 133, 134, 135, 114].

3.2.1 Transfer Learning Framework

Transfer learning is crucial for EEG emotion recognition because collecting extensive labeled calibration data for every new user is impractical, and EEG distributions vary across subjects and sessions even under identical elicitation protocols. Surveys consistently emphasize that subject-independent performance depends on explicit mechanisms to mitigate covariate/representation shift and negative transfer, especially under cross-session and cross-dataset deployment constraints [115, 114, 135]. Transfer learning in this context includes (i) *domain adaptation* (same task, shifted distributions), (ii) *domain generalization* (generalize to unseen domains without target data), and (iii) *fine-tuning* (calibration-efficient parameter adaptation), often combined in hybrid pipelines.

Domain Adaptation A persistent obstacle to *subject-independent* EEG emotion recognition is domain shift, i.e., the mismatch between the training distribution (source domain) and the deployment distribution (target domain). In affective EEG, a “domain” typically corresponds to a *subject*, *session/day*, or *dataset/device montage*, and domain shift arises from strong inter-subject variability, nonstationarity, and acquisition differences. Domain adaptation (DA) addresses this challenge by learning representations and decision rules that remain discriminative for emotion labels while reducing sensitivity to domain-specific cues [136, 137, 138].

In practical EEG emotion systems, models are often trained on multi-subject datasets but must generalize to unseen users under leave-one-subject-out (LOSO) or cross-session protocols. However, EEG distributions can vary substantially across subjects and sessions even under the same task, causing performance drops when optimizing only the source empirical risk. Formally, with labeled source data $\mathcal{D}_s = (x_i^s, y_i^s)_{i=1}^{n_s}$ and (often unlabeled) target data $\mathcal{D}_t = x_j^t, j = 1^{n_t}$, domain shift implies $P_s(X, Y) \neq P_t(X, Y)$, which may reflect covariate shift ($P(X)$), label shift ($P(Y)$), or conditional shift ($P(Y|X)$). DA mitigates these effects by incorporating an *alignment* term into training so that features extracted from source and target become closer under a chosen discrepancy while retaining emotion-relevant structure [30, 29, 19]. Domain adaptation (DA) addresses $P_s(X, Y) \neq P_t(X, Y)$ by learning representations that remain emotion-discriminative while being domain-invariant. With labeled source data $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ and target data $\mathcal{D}_t = \{x_j^t\}_{j=1}^{n_t}$ (often unlabeled), many DA methods optimize:

$$\min_{\theta_f, \theta_c} \mathcal{L}_{\text{cls}}(\mathcal{D}_s; \theta_f, \theta_c) + \lambda \mathcal{L}_{\text{align}}(\mathcal{D}_s, \mathcal{D}_t; \theta_f), \quad (5)$$

where $f_{\theta_f} : x \mapsto h \in \mathbb{R}^d$ is a feature extractor, θ_c parameterizes the emotion classifier, $\mathcal{L}_{\text{cls}}$ is supervised label loss on source, and $\mathcal{L}_{\text{align}}$ reduces source–target feature discrepancy.

Beyond single-source settings, Chen *et al.* introduced MS-MDA for *cross-subject and cross-session* EEG emotion recognition, using a shared extractor for domain-invariant features plus *multiple one-to-one adaptation branches* (each source paired with the target) to reduce negative transfer; they reported improvements on SEED/SEED-IV transfer protocols, while the approach increases architectural/training complexity and depends on stable normalization choices across domains [139]. To reduce the adaptation cost, Zhao *et al.* proposed a plug-and-play domain adaptation strategy that can adapt a trained model to a new subject with minimal modification, reporting gains in cross-subject emotion recognition; a limitation is that performance may degrade when target shifts are large or when pseudo-labels are noisy [124]. For parameter efficiency, Zhang *et al.* proposed PARSE, a parameter-efficient DA model for EEG emotion recognition that aims to adapt across subjects with fewer tunable parameters; such efficiency is attractive for deployment, but careful tuning is still needed to avoid under/over-adaptation [109]. Multi-source DA has also progressed: Li *et al.* proposed MS-FRAN to leverage multiple source domains with improved transferability, showing stronger cross-subject performance, but with added training overhead and sensitivity to heterogeneous source quality [130].

Graph-based encoders have been combined with DA to better respect EEG channel topology. For example, Xu *et al.* proposed DAGAM, a domain-adversarial graph attention model, reporting improvements for subject-independent emotion recognition by jointly modeling spatial relations and domain invariance; limitations include computational cost and the need for stable graph construction across datasets/montages [127]. Representation learning objectives that explicitly target invariance are also influential: Shen *et al.* [33] learned *subject-invariant* representations via contrastive learning for cross-subject EEG emotion recognition, often improving robustness without requiring a strong discriminator, but performance can depend on augmentation design and batch composition. For *cross-dataset* settings, Imtiaz & Khan studied unsupervised domain adaptation to enhance cross-dataset EEG emotion recognition and reported gains under dataset/device mismatch; remaining challenges include montage differences and residual conditional shift that cannot be solved by marginal alignment alone [140]. Earlier domain-alignment baselines for affective EEG, such as latent-representation similarity objectives, remain relevant as lightweight alternatives, but may underperform when shifts are highly nonlinear [126]. Related empirical work on cross-corpus/cross-dataset affect recognition further supports that domain mismatch is a primary error source, motivating stronger DA and domain generalization strategies [32, 132, 141].

Discrepancy-Based Methods Discrepancy-based domain adaptation methods aim to reduce the statistical divergence between source and target feature distributions in order to enable knowledge transfer across domains. In EEG-based emotion recognition, this approach is particularly relevant because physiological signals vary significantly across subjects, sessions, and recording environments, causing domain shifts that degrade model generalization. To address this issue, discrepancy-based methods explicitly measure and minimize distribution differences between domains using metrics such as Maximum Mean

Discrepancy (MMD) and Correlation Alignment (CORAL). By aligning feature representations in a shared latent space, these techniques encourage the learned representations to become domain-invariant while preserving discriminative information for emotion classification.

Discrepancy-based DA minimizes explicit distribution distances (e.g., MMD/CORAL). Li *et al.* [142] did cross-subject EEG emotion recognition by applying deep discrepancy alignment (DAN with multi-layer distribution matching) and reported accuracy improvements of **4.62%** (3-class) and **4.24%** (4-class) over baseline methods on SEED/SEED-IV; limitation: marginal discrepancy minimization may under-align nonlinear or class-conditional shifts and requires careful layer/feature selection. Bao *et al.* [143] did two-level DA by pairing discrepancy alignment (MMD) with adversarial confusion and normalization mechanisms, reporting strong cross-subject accuracy of **87.9%** on SEED and cross-day transfer accuracy of **74.93%** on SEED; limitation: multi-stage losses and topology mapping increase pipeline complexity and can be sensitive to scheduling. Chen *et al.* [144] did multi-source marginal distribution adaptation (MS-MDA) with per-source branches (one-to-one adaptation) and reported cross-subject accuracy of **89.63±6.79** on SEED (and **59.34±5.48** on SEED-IV), outperforming single-source baselines; limitation: computation and memory scale with the number of sources, and normalization stability across domains is required.

Overall, discrepancy-based domain adaptation methods attempt to improve cross-subject EEG emotion recognition by explicitly minimizing distribution mismatches between domains. Approaches such as deep adaptation networks, multi-level discrepancy alignment, and multi-source marginal distribution adaptation have demonstrated improved classification performance on benchmark datasets like SEED and SEED-IV. However, these methods still face several challenges, including sensitivity to feature-layer selection, increased computational complexity in multi-source settings, and difficulty in fully capturing nonlinear or class-conditional distribution shifts across domains.

Adversarial-Based Methods Adversarial domain adaptation (DA) approaches aim to learn domain-invariant representations by introducing an adversarial training mechanism between a feature extractor and a domain discriminator. In EEG-based emotion recognition, adversarial methods are widely adopted to mitigate cross-subject and cross-session variability, which often causes significant domain shifts between training and testing data. These approaches draw inspiration from generative adversarial networks (GANs), where the feature extractor attempts to generate representations that confuse the domain discriminator, thereby encouraging the model to learn features that are indistinguishable across domains. By aligning feature distributions implicitly through adversarial learning, these methods can capture complex nonlinear domain relationships that may not be effectively handled by explicit discrepancy minimization techniques.

Adversarial DA uses a domain discriminator to enforce invariance. Jin *et al.* [145] did EEG emotion recognition by adversarial domain adaptation and reported improved LOSO performance compared with non-adaptive baselines; limitation: early single-source adversarial alignment can be sensitive to heterogeneous subjects and may not fully handle conditional mismatch. Luo *et al.* [146] did adversarial transfer using a Wasserstein GAN objective (with gradient penalty) and showed improved stability and cross-subject performance on SEED/DEAP; limitation: adversarial training increases complexity and may require careful tuning to avoid instability. Zhao *et al.* [147] did calibration-efficient adversarial adaptation via plug-and-play DA by separating emotion-shared and subject-private components, achieving mean LOSO accuracy of about **86.7%** on SEED while reducing calibration to **within a minute**; limitation: still assumes a short target calibration stream and can degrade under large target shifts or pseudo-label noise. Xu *et al.* [127] did adversarial DA with a graph-attention encoder (DAGAM) to preserve channel topology while enforcing domain confusion, improving subject-independent recognition; limitation: graph construction and attention modules raise computation and sensitivity to montage mismatch. Yang *et al.* [148] did multi-source adaptation by aligning spectral-spatial attention patterns (attention-consistent alignment), improving robustness under cross-subject/cross-session transfer; limitation: attention alignment adds model complexity and may overfit under small targets.

Domain Adversarial Networks (DANN) A canonical DANN objective combines label prediction

with an adversarial domain confusion term:

$$\min_{\theta_f, \theta_c} \max_{\theta_d} \mathcal{L}_{\text{DANN}} = \underbrace{\frac{1}{n_s} \sum_{i=1}^{n_s} \ell(C_{\theta_c}(F_{\theta_f}(x_i^s)), y_i^s)}_{\mathcal{L}_{\text{cls}}} - \lambda \underbrace{\left[\frac{1}{n_s} \sum_{i=1}^{n_s} \ell_d(D_{\theta_d}(F_{\theta_f}(x_i^s)), 0) + \frac{1}{n_t} \sum_{j=1}^{n_t} \ell_d(D_{\theta_d}(F_{\theta_f}(x_j^t)), 1) \right]}_{\mathcal{L}_{\text{dom}}}, \quad (6)$$

where F_{θ_f} is a feature extractor ($h = F(x) \in \mathbb{R}^d$), C_{θ_c} is an emotion classifier, D_{θ_d} is a domain discriminator, and $\ell(\cdot), \ell_d(\cdot)$ are label and domain losses (often cross-entropy). The trade-off λ controls how aggressively domain cues are suppressed (via gradient reversal) while emotion discriminability is maintained [136, 133]. In practice, strong DANN-style alignment is often paired with normalization/discrepancy losses to stabilize training and prevent class collapse, as in TDANN and related hybrids [143, 146].

Overall, adversarial domain adaptation techniques aim to learn domain-invariant EEG representations by training a feature extractor to confuse a domain discriminator while preserving emotion-discriminative features. Methods ranging from early adversarial alignment models to more advanced architectures incorporating Wasserstein objectives, graph-attention encoders, and multi-source attention alignment have demonstrated improved subject-independent emotion recognition performance. Despite these advances, adversarial approaches still face challenges such as training instability, increased computational complexity, sensitivity to domain heterogeneity, and reliance on calibration data or pseudo-label quality in certain scenarios.

Prototype/Cluster-Based Methods Prototype or cluster-based domain adaptation methods aim to preserve class-discriminative structure while aligning source and target domains. Instead of only matching marginal distributions, these approaches typically construct class prototypes or cluster centers and align target samples using pseudo-labels or similarity measures. Such strategies are useful in EEG emotion recognition where class structure may shift across subjects, making purely distribution-based alignment insufficient.

Prototype/cluster-based DA preserves class structure via pseudo-labeling and prototype alignment. Peng *et al.* [149] did class-aware DA by joint feature adaptation and graph adaptive label propagation for cross-subject emotion recognition, improving performance by leveraging graph-structured pseudo-label relations; limitation: label propagation can amplify early pseudo-label noise if target confidence is poor. Shen *et al.* [33] did subject-invariant representation learning using contrastive objectives for cross-subject emotion recognition, often improving robustness without requiring a strong discriminator; limitation: gains depend on augmentation policy, batch composition, and the stability of positive/negative pair construction.

Overall, prototype and cluster-based approaches aim to maintain class consistency during adaptation by using pseudo-labels, graph propagation, or contrastive clustering mechanisms. These strategies often complement discrepancy or adversarial alignment by explicitly preserving class structure, though their performance can be sensitive to pseudo-label quality and representation stability.

Source-Free Adaptation Source-free domain adaptation (SFDA) addresses scenarios where the original source data are unavailable during adaptation, often due to privacy, storage, or data-sharing restrictions. Instead of accessing source samples, SFDA methods adapt a pretrained source model directly using only target-domain data, typically through pseudo-labeling, self-training, or test-time adaptation mechanisms. Source-free DA adapts models using only the target data at adaptation time (often due to privacy or data sharing constraints). Imtiaz & Khan [150] did cross-dataset unsupervised target-side adaptation using gradual pseudo-target selection (GPTDS) and selective test-time augmentation (PC-TTA), reporting **67.44%** (DEAP→SEED) and **59.68%** (SEED→DEAP) with improvements of **7.09%** and **6.07%** over baselines and a **15×** reduction in test-time augmentation compute; limitation: cross-dataset accuracy remains substantially lower than within-dataset transfer and pseudo-sample selection can be brittle under extreme domain gaps.

Overall, source-free DA provides a practical solution when source datasets cannot be accessed during deployment. While approaches such as GPTDS demonstrate promising improvements for cross-dataset EEG emotion recognition, their performance still depends heavily on the reliability of pseudo-label selection and robustness to large domain discrepancies.

Domain Generalization Domain Generalization (DG) aims to learn models that can generalize to unseen domains without using target-domain data during training. In EEG emotion recognition, DG is particularly important because EEG signals vary significantly across individuals, making subject identity a major source of domain shift. Therefore, DG methods attempt to learn invariant representations across multiple source domains to reduce subject-specific bias and improve cross-subject generalization [132, 115].

Zhou et al. [132] surveyed domain generalization methods by analyzing invariant representation learning techniques such as meta-learning, feature alignment, and data augmentation, but their study mainly focused on computer vision applications with limited coverage of physiological signals like EEG. Apicella et al. [115] investigated domain generalization strategies for neural signal analysis by reviewing multi-source training and invariant feature learning methods, but they reported that large inter-subject variability in EEG still limits generalization performance. Wang et al. [14] studied EEG-based emotion recognition using temporal, spatial, and frequency-domain feature learning techniques to improve cross-subject performance, but their approach remains sensitive to dataset-specific characteristics and lacks explicit domain-invariant learning mechanisms. Li et al. [151] explored deep learning models such as CNN and RNN architectures for EEG emotion recognition by learning hierarchical representations from raw signals, but these models often overfit to subject-specific patterns and struggle to generalize across domains. Overall, DG methods help improve cross-subject robustness in EEG emotion recognition by reducing subject-specific bias. Invariant representation learning works well when multiple source domains are available, while augmentation-based strategies are useful when training data are limited.

Fine-Tuning Strategies Fine-tuning strategies adapt pretrained models to a target domain using a small labeled calibration set. In EEG emotion recognition, fine-tuning is commonly used when limited labeled samples from a new subject or session are available. By updating only a subset of parameters or applying regularization during training, fine-tuning enables the model to retain previously learned knowledge while adapting to the specific characteristics of the target domain. Fine-tuning adapts a pretrained encoder/classifier to the target with a small labeled calibration set. A regularized fine-tuning objective is:

$$\min_{\theta} \mathcal{L}_{\text{FT}} = \frac{1}{|\mathcal{D}_t^\ell|} \sum_{(x,y) \in \mathcal{D}_t^\ell} \ell(f_\theta(x), y) + \mu \|\theta - \theta_s\|_2^2, \quad (7)$$

where f_θ is the adapted model, $\mathcal{D}_t^\ell$ is a small labeled target set, θ_s are pretrained parameters, and μ regularizes updates to avoid overfitting. Quan *et al.* [152] did multi-source transfer with MR-VAE representation learning and explicit source-domain selection, reporting **92.83%** (SEED) and **79.30%** (SEED-IV); limitation: requires a small labeled target set and can still suffer negative transfer if domain selection is imperfect. Lu *et al.* [153] did hybrid fine-tuning and DA in DFF-Net and reported **93.37%** (SEED) and **82.32%** (SEED-IV), with ablations showing fine-tuning alone outperformed DA alone on SEED; limitation: needs target samples and was noted to lack online real-time capability. Wang *et al.* [154] did cross-dataset transfer by pretraining a ViT encoder (PESD) and adapting with Mixup, triplet loss, and a domain discriminator, achieving strong accuracies on multiple datasets (e.g., **93.14%** on SEED and **93.53%** on DEAP); limitation: large pretraining and compound losses increase training complexity and compute requirements.

Across subject-independent EEG emotion recognition, **multi-source DA** (e.g., MS-MDA and attention-aligned MSDA) is typically best when many source subjects are available because it reduces negative transfer by respecting per-source heterogeneity [144, 148]. **Discrepancy-based hybrids** (e.g., TDANN

combining MMD and adversarial confusion) often offer strong within-dataset cross-subject accuracy but require careful tuning and can be complex [143]. **Calibration-efficient adversarial methods** (e.g., PPDA) are preferable when user burden is critical, trading some peak accuracy for rapid adaptation [147]. Under severe **cross-dataset** mismatch, source-free unsupervised adaptation and pretrained encoders are increasingly competitive, though absolute accuracy remains lower than within-dataset transfer and robustness hinges on pseudo-sample reliability and augmentation design [150, 154].

Overall, fine-tuning strategies remain an effective mechanism for adapting pretrained EEG emotion recognition models to new subjects or datasets when a small labeled calibration set is available. Recent works combining fine-tuning with multi-source transfer, domain adaptation, or large-scale pretraining have achieved strong cross-subject and cross-dataset performance. However, these approaches still depend on limited labeled target data and may introduce additional training complexity, especially when hybrid adaptation frameworks or large pretrained encoders are employed.

3.3 Machine Learning and Deep Learning Methods

Machine learning and deep learning methods define EEG emotion recognition models through three core design decisions: the representation that converts raw EEG into learnable inputs, the hypothesis class that maps representations to affect labels, and the optimization strategy that estimates parameters under limited and noisy supervision [155]. The model family also implies an assumption about what remains stable across trials and subjects, because shallow models often assume engineered features already encode invariances [156], whereas deep models often learn invariances from data through inductive bias and regularization [157]. A practical description of “the model” therefore includes not only the classifier but also the feature extraction operator, the regularizers that control complexity, and the training protocol that prevents leakage-driven shortcuts in time-windowed EEG segmentation [158]. Model choice also reflects an implicit stance on nonstationarity, because feature-engineering pipelines often treat stationarity as a preprocessing outcome [159], while end-to-end learners treat stationarity as a representation-learning target [160]. The model specification, therefore, remains inseparable from the data collection and annotation process, because stimulus design and subject instruction define the distribution that the optimizer actually sees [161].

Machine learning and deep learning methods formalize EEG emotion recognition as supervised prediction from multichannel observations to affect labels, so the methodological pipeline couples signal representation with a discriminative decision rule [156]. A generic formulation defines an EEG segment as $\mathbf{X} \in \mathbb{R}^{C \times T}$ with C channels and T samples, and it defines a model $f_\theta(\cdot)$ that outputs a class posterior $\hat{\mathbf{p}} \in \Delta^{K-1}$ over K affect classes. The learning objective commonly minimizes empirical risk with regularization,

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N \mathcal{L}(f_\theta(\mathbf{X}_i), y_i) + \lambda \|\theta\|_2^2, \quad (8)$$

where $\mathcal{L}$ often uses cross-entropy $\mathcal{L}_{\text{CE}} = -\sum_{k=1}^K \mathbb{1}[y = k] \log \hat{p}_k$. Review syntheses emphasize that methodological validity depends on protocol control, because windowing, overlap, and subject partitioning determine whether θ encodes emotion-relevant structure or leakage-driven shortcuts [160]. The same formulation also implies a modeling choice about label semantics, because discrete emotion categories, dimensional bins, and intensity scales define different decision boundaries even when the same architecture applies [157, 161]. Label noise and class overlap, then shift the practical goal from “perfect classification” toward robust risk minimization under ambiguous supervision [162]. Confidence calibration and error auditing, therefore, remain essential alongside accuracy metrics [163, 164]. The training objective also interacts with sample construction, because heavy overlap in sliding windows increases dependence between training and test examples unless partitioning rules remain strict [158].

Representation choice governs model capacity requirements, so shallow pipelines typically construct explicit feature vectors $\mathbf{z} = \phi(\mathbf{X})$ and train a classical classifier $g(\mathbf{z})$ [165], while deep pipelines learn $\phi_\theta(\cdot)$ jointly with $g_\theta(\cdot)$ [155]. Time-frequency encoders often use short-time Fourier transforms (STFT) or wavelet-

like decompositions, for example,

$$\text{STFT}_{\mathbf{x}}(t, \omega) = \sum_{\tau} \mathbf{x}[\tau] w[\tau - t] e^{-j\omega\tau}, \quad (9)$$

and band-limited power features often use Welch-style spectra $P(f)$ and band energy $E_b = \int_{f \in b} P(f) df$. Deep models often bypass explicit feature design by learning convolutional, recurrent, attention, or graph operators that map $\mathbf{X}$ to latent embeddings $\mathbf{h}$ that preserve spatial topology and temporal context [166]. This dichotomy also influences computational cost and deployment feasibility, because shallow methods shift effort toward offline feature engineering [167], while deep methods shift effort toward training stability and hardware acceleration [164]. A practical system often mixes these paradigms by retaining signal transformations that stabilize nonstationarity and then applying learnable models on top [168]. Artifact-aware automation further reduces brittleness under session drift [64]. Feature normalization and subject-wise standardization also matter because EEG amplitude scales differ across individuals and electrodes, and inconsistent scaling undermines both margin-based classifiers and gradient-based learners [156].

Evaluation also shapes methodological emphasis, because subject-dependent settings reward personalization [156], while subject-independent settings reward invariance under inter-subject shift [169]. A unified view defines a training distribution $P_S(\mathbf{X}, y)$ and a target distribution $P_T(\mathbf{X}, y)$ and measures generalization under $P_T \neq P_S$, so robust methods constrain representation drift and reduce discrepancy,

$$\mathcal{R}_T(f) \leq \mathcal{R}_S(f) + \text{disc}(P_S, P_T) + \epsilon, \quad (10)$$

where $\text{disc}(\cdot, \cdot)$ denotes a divergence proxy. This viewpoint motivates explicit shift-handling objectives and transfer strategies [170]. Subject-specific covariates often dominate cross-subject error [171], while low-dimensional transfer formulations provide alternative routes to generalization [172]. Reporting practices matter because accuracy averages conceal instability across subjects, so protocol-aware analysis benefits from per-subject breakdowns and variance-aware summaries [158]. Cross-dataset evaluation further intensifies these concerns, because device differences and preprocessing heterogeneity amplify distribution shift beyond typical inter-subject variability [173].

3.3.1 Shallow Models

Shallow models in EEG emotion recognition rely on an *explicit* feature map $\phi(\cdot)$ that converts the raw signal $\mathbf{X}$ into a moderate-dimensional, engineered representation $\mathbf{z} = \phi(\mathbf{X})$, after which a classical classifier learns a relatively simple decision rule. Because the inductive bias is primarily encoded in the hand-crafted descriptors (e.g., time–frequency statistics, bandwise entropy, connectivity), shallow pipelines remain appealing when labels are limited and interpretability is desired, since each feature dimension can often be linked to a physiological hypothesis and inspected across trials or subjects [159, 174, 175, 167]. Liu *et al.* did shallow affect modeling by emphasizing an explicit feature map and compact affect-sensitive descriptors (e.g., entropy-style band statistics) with the limitation that performance depends strongly on preprocessing assumptions and hand-engineering choices [159]. Differential-entropy (DE) features treat band-limited EEG as approximately Gaussian with variance σ_b^2 so that

$$h_b = \frac{1}{2} \log! (2\pi e, \sigma_b^2), \quad (11)$$

yielding compact descriptors that are widely used in shallow affect modeling [159, 176]. Lieta *et al.* did EEG emotion recognition by organizing the pipeline around feature-family specification and feature selection with the limitation that the selected subset can be unstable under session drift and may not preserve nonlinear class structure [174]. Feature selection is commonly posed as a sparsity-regularized regression problem

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{Z}\mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_1, \quad (12)$$

where $\mathbf{Z}$ stacks engineered features, $\mathbf{y}$ denotes labels (or label-coded targets), and $\mathbf{w}$ indicates feature relevance. Vee *et al.* did shallow classification by selecting a classifier geometry matched to the engineered feature distribution (e.g., margin-based decision rules) with the limitation that classifier performance is bounded by the separability of $\phi(\mathbf{X})$ and may degrade under cross-subject variability [176]. Hadi *et al.* did a survey-style synthesis of interpretable handcrafted EEG emotion features and classical classifiers with the limitation that the reported comparisons are often confounded by inconsistent preprocessing and evaluation splits [175]. Zhou *et al.* did wavelet/transform-driven feature extraction with classical learners and ensembles with the limitation that time–frequency decompositions introduce design degrees of freedom (mother wavelet, levels, statistics) that can be dataset-specific [167]. In many such transform-based pipelines, decomposition-driven representations express each EEG channel as a sum of intrinsic mode functions (IMFs) plus a residual term,

$$x_c(t) = \sum_{m=1}^M \text{IMF}_{c,m}(t) + r_c(t), \quad (13)$$

from which per-IMF or per-band statistics are computed for classification. Zhang *et al.* did spatial/connectivity-informed handcrafted representation learning (inter-channel coupling features) with the limitation that connectivity estimates can be noisy and sensitive to artifacts, reference choice, and short window lengths [177]. Lai *et al.* did artifact-aware shallow EEG pipelines by highlighting how ocular and muscular contamination distorts engineered descriptors with the limitation that incomplete artifact suppression can bias feature importance and downstream selection [178]. Sain *et al.* did automated preprocessing/cleaning to support repeatable feature extraction with the limitation that automation may still fail on atypical artifacts and can propagate systematic errors into features [168]. Hadi *et al.* did automatic feature/channel selection to increase robustness (e.g., stable subsets of channels/bands) with the limitation that selection can overfit to nuisance-correlated dimensions when artifacts are residual [64]. Wank *et al.* did interpretable shallow classification emphasizing stable decision rules (e.g., SVM/ k -NN family) with the limitation that hyperparameter sensitivity (e.g., C , distance metrics) can be high when features are redundant or unscaled [165]. Witte *et al.* did comparative evaluation emphasizing calibration and emphasizing that miscalibrated confidence can undermine downstream affect-aware decisions with the limitation that calibration quality can vary substantially across classifiers and datasets even at similar accuracies [162]. Egger *et al.* did the use of shallow baselines (feature engineering + classical learners) as transparent reference points with the limitation that careful coupled design of cleaning, selection, and classification is needed to avoid fragile performance under subject/session variability [160].

Overall, shallow models are typically best when *interpretability*, *small labeled data*, or *tight compute budgets* dominate: compact band-wise descriptors (e.g., DE) paired with stable classifiers provide strong, transparent baselines [159, 176]. Wavelet/decomposition-driven features and ensembles are often preferable when nonstationarity is prominent and variance reduction is needed, but they require careful design to avoid dataset-specific tuning [167]. Connectivity-informed handcrafted features are most useful when the research goal is to relate emotion to *distributed* coupling patterns, but they are especially sensitive to artifacts and preprocessing choices, so rigorous cleaning and stable selection are prerequisites [177, 178, 64]. Finally, when confidence is used downstream (e.g., affect-aware control), calibration-aware evaluation becomes critical even for shallow pipelines, since unreliable probabilities can be harmful despite high accuracy [162, 160].

3.3.2 Deep Learning Models

Deep learning models address EEG emotion recognition through *learnable* feature extraction $\phi_\theta(\mathbf{X})$, making architecture and training constraints the primary determinants of what invariances are captured. This perspective is closely related to **domain generalization**: because EEG varies across subjects, sessions, and acquisition settings, deep models must learn representations that encode emotion-relevant structure while suppressing nuisance factors (e.g., subject identity, residual artifacts) that can dominate high-capacity learners. Consequently, the model description emphasizes inductive bias from architectural

families—convolution for locality, recurrence for temporal dynamics, and topology-aware encoders for inter-channel relations—and it emphasizes training procedures (augmentation, normalization, and regularization) that prevent memorization and improve out-of-domain behavior [155, 164, 166, 177, 179, 158]. Roy *et al.* did deep learning-based EEG affect modeling by emphasizing that architecture and protocol discipline determine whether models learn emotion or leak subject identity with the limitation that weak experimental control can inflate accuracy through leakage and bias [155]. Cheng *et al.* did CNN-based EEG emotion recognition by exploiting spatial–spectral locality with the limitation that convolutional filters may miss long-range temporal interactions unless combined with temporal modules [164]. Liu *et al.* did subject-independent emotion recognition by recurrent/sequential modeling to capture temporal dependence with the limitation that recurrent architectures can be computationally heavy and sensitive to sequence length and optimization issues [166]. Zhang *et al.* did topology-aware modeling by encoding inter-channel coupling (spatial/relational structure) with the limitation that relational priors and connectivity surrogates can be noisy and sensitive to preprocessing or montage differences [177]. Aydeniz *et al.* did deep EEG emotion learning by using augmentation/normalization/regularization to enforce invariance under nuisance variability with the limitation that poorly designed EEG augmentations may introduce unrealistic distortions or amplify spurious cues [179]. Suhaimi *et al.* did EEG emotion recognition benchmarking and analysis of deep learning practices with the limitation that inconsistent preprocessing and evaluation splits can induce annotation bias and optimistic performance, especially when segmentation leakage is present [158]. Egger *et al.* did comparative modeling by contrasting deep and shallow pipelines and highlighted that deep models can improve robustness under moderate noise once properly regularized, with the limitation that they require stricter protocol control and sufficient data to avoid overfitting [160]. Kim *et al.* did emotion modeling with deep autoencoder-style representation learning to reduce handcrafted dependence with the limitation that unsupervised features may absorb nuisance structure unless constrained by task supervision and explicit regularization [163]. Hadi *et al.* did automated preprocessing/selection to stabilize deep EEG training under high-dimensional inputs with the limitation that residual artifacts and channel quality variation can still bias learned features [64]. Lai *et al.* did artifact-aware EEG processing by emphasizing the impact of ocular/muscular contamination on downstream learning with the limitation that incomplete artifact removal can propagate noise into end-to-end representations [178]. He *et al.* did channel-attention EEG modeling to capture long-range channel interactions with the limitation that attention weights are not guaranteed to be causally interpretable and can reflect optimization shortcuts [180]. Tan *et al.* did a survey of affective computing methods that included attention-based deep models and interpretability tools with the limitation that saliency/attention visualizations can be unstable across runs and do not necessarily validate physiological causality [161].

Overall, CNN-centric models are often best when local spatial–spectral patterns dominate and training data are moderate, whereas RNN/sequential models are preferable when temporal dynamics carry most discriminative information but can be harder to optimize [164, 166]. Topology-aware encoders are most useful when cross-electrode coupling is central to the affective mechanism, although they can be sensitive to montage and preprocessing [177]. Attention/transformer-style models are best when long-range, context-dependent interactions are critical, but they demand careful regularization and protocol rigor to prevent leakage and spurious saliency [180, 158, 155]. Across all families, the strongest empirical lesson is that domain-generalizable deep EEG emotion recognition depends as much on disciplined evaluation and artifact-aware preprocessing as on architecture choice [178, 64, 160].

Standalone Models Standalone deep models learn a *single* end-to-end mapping from EEG inputs to emotion labels and are therefore a useful lens for **domain generalization**: they expose how much cross-subject/cross-session robustness can be obtained from architectural inductive bias and disciplined training alone, without additional fusion, multi-branch alignment, or auxiliary objectives. In EEG emotion recognition, standalone baselines are valuable because they isolate representation learning in $\phi_{\theta}(\mathbf{X})$ and reduce confounding factors (e.g., heterogeneous feature fusion and branch-specific preprocessing), making results easier to reproduce and compare across protocols [179, 160]. Their model specification typi-

cally centers on backbone type (CNN/RNN), input encoding (raw, time–frequency, or spatial maps), and depth–width trade-offs that control expressivity versus overfitting and subject memorization [164]. Aydeniz *et al.* did standalone deep EEG emotion learning by emphasizing augmentation/normalization/regularization to enforce invariance under nuisance variability with the limitation that augmentations can inject unrealistic distortions if not EEG-aware [179]. Cheng *et al.* did standalone CNN-based recognition by exploiting spatial–spectral locality through convolutional backbones with the limitation that purely local kernels can miss long-range temporal dependencies unless complemented by temporal modeling [164]. Liu *et al.* did standalone sequential modeling by using recurrent backbones to capture temporal dependence with the limitation that recurrence can be harder to optimize and computationally heavier than purely convolutional designs [166]. Egger *et al.* did reproducible benchmarking by comparing deep standalone models against feature-engineered baselines and highlighted that standalone deep models can be strong under controlled protocols with the limitation that performance can be inflated by leakage or optimistic splitting if protocol discipline is weak [160]. Suhaimi *et al.* did empirical evaluation/benchmarking of EEG emotion recognition pipelines and emphasized that a single backbone reduces confounds from heterogeneous preprocessing with the limitation that inconsistent dataset preprocessing and evaluation splits still dominate reported variance across papers [158]. Wank *et al.* did efficiency-oriented modeling by emphasizing latency- and memory-aware classifiers (often used as real-time baselines) with the limitation that lightweight models may underfit complex EEG patterns unless input encoding and regularization are carefully chosen [165].

A generic standalone convolutional block can be summarized as

$$\mathbf{H}^{(l)} = \sigma(\mathbf{W}^{(l)} * \mathbf{H}^{(l-1)} + \mathbf{b}^{(l)}), \quad (14)$$

where $\mathbf{H}^{(l-1)}$ and $\mathbf{H}^{(l)}$ are the input/output feature maps at layer l , $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are trainable weights and biases, $*$ denotes convolution over the chosen EEG tensorization (e.g., channel×time or time–frequency maps), and $\sigma(\cdot)$ is a nonlinearity. CNN-only standalone models are attractive because they match spatial locality in EEG topographies and time–frequency representations, and they avoid multi-branch fusion instabilities, which can otherwise obscure whether gains come from architecture or from fusion heuristics [166].

Overall, standalone models are best when the goal is **clean benchmarking** and **ablation-driven insight**: they provide the most interpretable evidence of what inductive bias (CNN vs. RNN) and what training protocol contribute to performance [160, 158]. CNN-only standalones are typically preferred when computations must be efficient and when discriminative cues are localized in time–frequency or spatial maps [164], while RNN-family standalones are preferred when long-range temporal dynamics are central but can be slower and harder to optimize [166]. For **real-time** affective systems with strict latency/memory constraints, lightweight standalone designs remain the practical choice, though they require careful regularization and input encoding to avoid underfitting [165].

Attention-Enhanced Models Attention-enhanced models extend deep EEG emotion recognition by making *relevance* a learnable function of the input, rather than a fixed property of particular channels or time points. This design choice is closely tied to **domain generalization**: because emotion-relevant EEG evidence can appear sparsely in time and vary in spatial distribution across subjects, attention provides a content-adaptive mechanism to emphasize discriminative substructures while suppressing nuisance variability. Consequently, attention-based model descriptions emphasize (i) the *granularity* of attention (channel-wise, temporal, or joint), (ii) the *scoring* function that produces weights, and (iii) the *integration rule* that mixes attended features with a backbone representation, since these choices determine whether the model learns robust affective cues instead of memorizing subject-specific patterns [180, 181, 182, 155].

He *et al.* did channel-attention EEG decoding by learning content-adaptive channel weighting to emphasize salient electrodes with the limitation that attention weights alone do not guarantee causal interpretability and can reflect optimization shortcuts [180]. Wei *et al.* did attention-based feature learning by integrating channel-wise weighting into a deep backbone to effectively down-weight noisy sen-

sors with the limitation that the learned attention can be sensitive to pooling design and regularization strength [181]. Xing *et al.* did adaptive attention modeling by learning dynamic reweighting across channels (and/or time) to boost signal-to-noise ratio for emotion recognition with the limitation that attention modules introduce extra parameters and may overfit under limited labels or weak evaluation protocols [182]. Zhang *et al.* did spatial/relational EEG modeling with temporal relevance mechanisms that highlight informative segments when affect evolves gradually, with the limitation that connectivity-driven representations and temporal weighting are sensitive to artifacts, referencing, and window length choice [177]. Roy *et al.* did deep EEG benchmarking and emphasized that attention-enhanced models (like other high-capacity designs) can amplify leakage and protocol bias unless training schedules and evaluation splits are strictly controlled, with the limitation that unstable optimization and weak protocol discipline can yield inflated performance [155].

Transformer-style attention increasingly appears because emotion-relevant cues exhibit long-range interactions across time and channels [180]. Self-attention defines queries, keys, and values as $\mathbf{Q} = \mathbf{H}\mathbf{W}_Q$, $\mathbf{K} = \mathbf{H}\mathbf{W}_K$, and $\mathbf{V} = \mathbf{H}\mathbf{W}_V$, and it computes

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right) \mathbf{V}, \quad (15)$$

which yields data-adaptive aggregation over tokens that represent channels, time steps, or time-frequency patches. Attention mechanisms fit EEG decoding because relevance concentrates in subsets of electrodes and epochs, so learned weighting acts as a soft selection operator that complements convolution or recurrence [166]. Attention-based designs also support post hoc analysis via saliency-like maps, yet interpretability remains conditional because weight patterns do not necessarily imply causal importance [161].

Attention-enhanced models introduce adaptive reweighting across channels and time, so they prioritize salient electrodes and temporal segments while suppressing irrelevant activity. Channel-wise attention often learns a gating vector $\boldsymbol{\alpha} \in [0, 1]^C$ and produces a reweighted signal $\tilde{\mathbf{X}} = \text{diag}(\boldsymbol{\alpha})\mathbf{X}$, where

$$\boldsymbol{\alpha} = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \text{pool}(\mathbf{X}))), \quad (16)$$

and $\text{pool}(\cdot)$ aggregates across time. Channel attention also acts as an implicit sensor-selection mechanism, which reduces sensitivity to noisy electrodes and facilitates reduced-montage systems [181, 182]. Temporal attention highlights salient segments within a trial, which matters when affect evolves gradually rather than instantaneously [177]. Attention-enhanced models also introduce optimization sensitivity, so they require careful learning rate schedules and regularization to avoid unstable convergence [155]. Overall, attention-enhanced models are best when discriminative evidence is **sparse** or **subject-varying** in space/time and when robustness benefits from emphasizing salient substructures. Channel-attention mechanisms are most useful when sensor noise or electrode relevance varies (enabling soft channel selection), whereas temporal attention is most useful when emotions evolve over time and informative segments are localized [181, 182, 177]. However, attention modules increase optimization sensitivity and can overfit in small-data settings, so they are most reliable under careful regularization, disciplined evaluation splits, and protocol checks that reduce leakage and annotation bias [155].

Hybrid Models Hybrid deep models for EEG emotion recognition combine multiple operator families (e.g., convolution, recurrence, attention, graph/relational modules) to capture complementary structure across space, time, and frequency. From a **domain generalization** perspective, hybridization is motivated by the fact that emotion-relevant EEG cues can shift in their dominant form across subjects and sessions (e.g., some subjects exhibit discriminative spatial patterns, others exhibit stronger temporal dynamics). Thus, hybrid models assume that no single inductive bias is universally sufficient and instead aim to learn representations that remain robust under nuisance variability by integrating multiple views of the signal [166]. The model specification therefore emphasizes the participating modules, the fusion stage, and coordination constraints (normalization, scale matching, and regularization) needed to stabilize training and prevent a single branch from dominating [163, 160].

Liu *et al.* did hybrid modeling by coupling sequential operators with learned feature extraction to capture long-range temporal dependence with the limitation that recurrent components increase optimization difficulty and computational cost, especially under long EEG sequences [166]. Kim *et al.* did hybrid representation learning by combining deep autoencoder-style feature learning with supervised emotion decoding to reduce handcrafted dependence with the limitation that unsupervised components may absorb nuisance structure unless constrained by task supervision and careful regularization [163]. Zhang *et al.* did hybrid spatial/relational modeling by combining spatial structure (connectivity/topology features) with temporal processing to encode distributed affective coupling, with the limitation that connectivity-driven representations can be noisy and sensitive to artifacts, referencing, and windowing [177]. He *et al.* did hybrid attention-enhanced decoding by integrating channel-wise attention with a backbone representation to emphasize salient electrodes and epochs, with the limitation that attention patterns are not guaranteed to be causally interpretable and may reflect optimization shortcuts [180]. Suhaimi *et al.* did empirical benchmarking of deep EEG emotion pipelines and highlighted that hybrid model gains can be confounded by inconsistent preprocessing and evaluation splits, with the limitation that reported improvements may partially reflect protocol artifacts rather than architecture alone [158]. Egger *et al.* did comparative evaluation and emphasized that hybridization increases expressivity but also introduces failure modes such as branch dominance and brittleness to representation choices, with the limitation that hybrids require disciplined ablations to attribute improvements to specific modules [160]. Li *et al.* did physiological-signal-driven hybrid modeling by integrating complementary modules to capture multi-scale affect dynamics, with the limitation that modular synchronization and optimization sensitivity can complicate reproducibility and cross-paper comparability [157].

CNN–RNN hybrids commonly pass convolutional embeddings $\mathbf{h}_t$ (local spatial/spectral features) into a gated recurrent unit (GRU) to model longer temporal dependencies. A GRU update is:

$$\mathbf{z}_t = \sigma(\mathbf{W}_z \mathbf{h}_t + \mathbf{U}_z \mathbf{s}_{t-1}), \quad \mathbf{r}_t = \sigma(\mathbf{W}_r \mathbf{h}_t + \mathbf{U}_r \mathbf{s}_{t-1}), \quad (17)$$

$$\tilde{\mathbf{s}}_t = \tanh(\mathbf{W} \mathbf{h}_t + \mathbf{U}(\mathbf{r}_t \odot \mathbf{s}_{t-1})), \quad \mathbf{s}_t = (1 - \mathbf{z}_t) \odot \mathbf{s}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{s}}_t, \quad (18)$$

where $\mathbf{s}_t$ is the hidden state, $\mathbf{z}_t$ and $\mathbf{r}_t$ are update/reset gates, and $\odot$ denotes element-wise multiplication. This structure encodes multi-scale context by pairing local feature extraction (CNN) with sequential memory (RNN/GRU), which is useful when affective dynamics evolve over extended time [177, 180]. Overall, hybrid models are best when emotion cues are **multi-factorial** (simultaneously spatial, temporal, and relational) and when the dataset/protocol provides enough coverage to train multi-module systems without overfitting. CNN–RNN (or CNN–Transformer) hybrids are often preferable when both short-term rhythmic patterns and longer temporal context are discriminative, whereas topology-aware+temporal hybrids are preferable when inter-channel coupling is central to affect representation [177, 180]. However, because hybrids introduce optimization sensitivity and branch-dominance risks, they are most reliable under careful normalization, regularization, and disciplined ablations that separate true architectural benefit from preprocessing or split artifacts [160, 158, 157].

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